

Introduction:

Filters are frequency-selective circuits used to pass or attenuate different frequency components of an electrical signal. In practical situations, a signal contains both useful and unwanted frequency components. A filter is used to reduce the unwanted components while retaining the frequency range of interest. An ideal filter does not change the frequencies already present in the input signal. Instead, it changes the relative amplitude and phase of the signal with respect to different frequency components.

This frequency-dependent behaviour of a filter is obtained by using reactive elements in the circuit. The impedance of a resistor is frequency-independent, whereas the impedances of a capacitor and an inductor change with frequency. Therefore, when resistors, capacitors and inductors are connected in a suitable arrangement, the voltages and currents in the circuit change with frequency. This property of frequency-dependent impedance of L and C is used to obtain the required filter response. Applications of filters involve noise reduction, removal of unwanted frequency components, signal smoothing, communication systems and many others.

Common Terminologies:

The response of a filter is generally studied in the frequency domain. Two quantities are used to describe the frequency-dependent behaviour of a filter: the **magnitude response** and the **phase response**. The magnitude response describes the change in amplitude with frequency, while the phase response gives the phase shift introduced by the filter.

Transfer Function: The transfer function of a filter describes the relationship between its output and input in the frequency domain. It is defined as the ratio of the output signal to the input signal under zero initial conditions,

$$H(s) = \frac{V_{\text{out}}(s)}{V_{\text{in}}(s)}.$$

Here, s is the complex frequency variable. For sinusoidal steady-state analysis, $s = j\omega$, and the frequency response is obtained as $H(j\omega)$. The magnitude $|H(j\omega)|$ gives the change in amplitude, while the phase of $H(j\omega)$ gives the phase shift introduced by the filter.

Passband and Stopband: The passband is the range of frequencies over which the signal is passed with relatively small attenuation, while the stopband is the range of frequencies over which the signal is significantly attenuated.

Transition Band: The transition band is the frequency range between the passband and stopband over which the response changes from the passed to the attenuated region.

Cutoff Frequency: The cutoff frequency is a characteristic frequency used to specify the boundary of the filter response. For many commonly used filter responses, it is defined as the frequency at which the magnitude of the response falls by 3dB from its passband value. For a voltage or current response, this corresponds to a ratio

$$\frac{|H(j\omega_c)|}{|H|_{\text{passband}}} = \frac{1}{\sqrt{2}} \approx 0.707.$$

Thus, for a unity-gain filter, the output amplitude at the cutoff frequency is approximately 0.707 times its passband value. However, the exact definition of cutoff frequency depends on the type of filter response being considered.

Decibel (dB): The decibel is a logarithmic unit used to express the ratio of two quantities relative to a reference value. For voltage gain,

$$G_{\text{dB}} = 20 \log_{10} \left| \frac{V_{\text{out}}}{V_{\text{in}}} \right|.$$

A value of 0 dB corresponds to equal input and output amplitudes, while a negative value indicates attenuation.

Filter Order: The order of a filter is the highest power of s appearing in the denominator of its transfer function and is equal to the number of poles of the filter. In many simple filter circuits, the order is also related to the number of independent reactive elements, such as capacitors and inductors. The order determines how rapidly the filter response changes beyond the cutoff region.

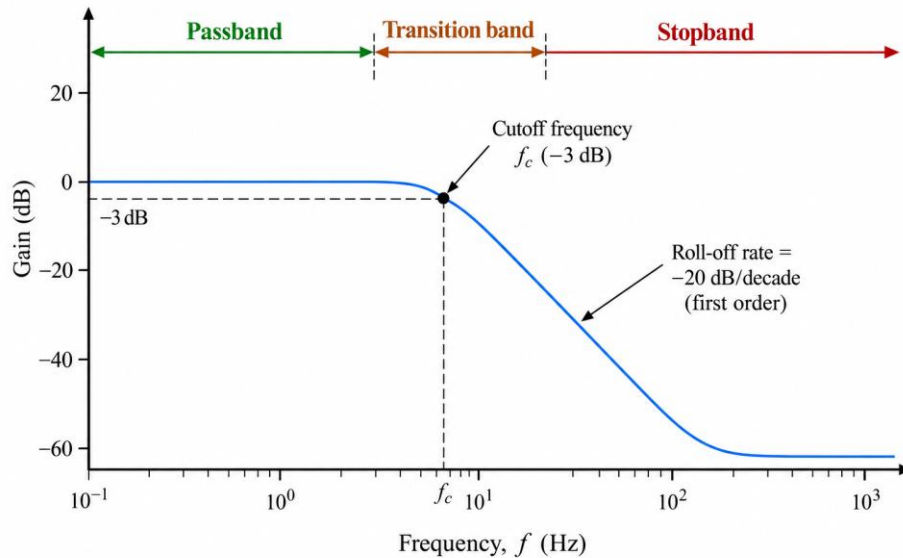
Roll-off Rate: The roll-off rate describes how rapidly the magnitude of the filter response decreases in the transition from the passband towards the stopband or vice-versa. It is commonly expressed in **dB/decade** or **dB/octave**. A decade represents a tenfold change in frequency. For example, 100 Hz $\rightarrow$ 1000 Hz represents one decade. An octave represents a twofold change in frequency. For example, 1 kHz $\rightarrow$ 2 kHz represents one octave.

For an n -th order filter, the asymptotic roll-off rate is approximately

$$20n \text{ dB/decade } \textit{or} \text{ } 6n \text{ dB/octave}.$$

Thus, a first-order filter has a roll-off of approximately 20dB/decade, while a second-order filter has a roll-off of approximately 40dB/decade.

The figure below shows a typical magnitude response of a filter that attenuates high frequencies and allows low frequency components to pass. Various related terminologies are shown through this example.



Classification of Filters:

Broad classification of filters is based on the components used to implement the filter.

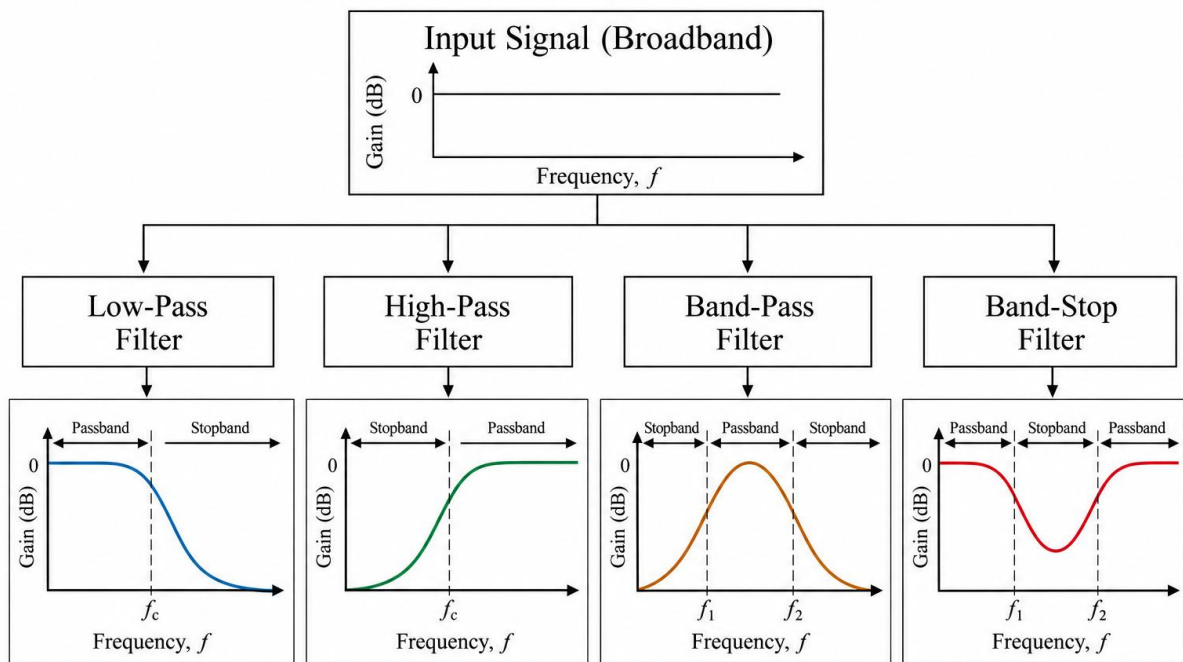
Passive filters are made using passive components such as resistors, capacitors and inductors. They do not require an external power supply for amplification and cannot provide power gain.

Active filters use active devices, most commonly operational amplifiers, together with resistors and capacitors. They can provide voltage gain and can also avoid some of the loading problems associated with passive networks. Active filters are commonly used for implementing higher-order filter circuits or when gain is needed in the output response.

This classification refers to the circuit implementation and does not determine the type of frequency response. Both passive and active filters can be designed to produce low-pass, high-pass, band-pass or band-stop responses. Based on their frequency-selective behaviour, filters are broadly classified as:

1. **Low-pass filter (LPF):** It allows lower-frequency components of a signal to pass while attenuates higher-frequency components. Its response is therefore high at low frequencies and decreases as the frequency increases beyond the cutoff region.
2. **High-pass filter (HPF):** It attenuates lower-frequency components while allowing higher-frequency components to pass. Its response increases through the transition region and approaches the passband value at higher frequencies.

3. **Band-pass filter (BPF):** It allows a particular range of frequencies to pass while attenuating frequencies below and above this range. It has a lower cutoff frequency f_L and an upper cutoff frequency f_H , with the bandwidth given by $B = f_H - f_L$. For a second-order band-pass response, the centre frequency is $f_0 = \sqrt{f_L f_H}$.
4. **Band-stop filter (BSF):** It attenuates a particular range of frequencies while allowing frequencies outside this range to pass. When the rejected frequency range is narrow and the main purpose is to suppress a particular unwanted frequency, it is commonly referred to as a **notch filter**. It also has two cutoff frequencies as in BPF.



Experiment 1: First-Order Passive Low Pass Filter

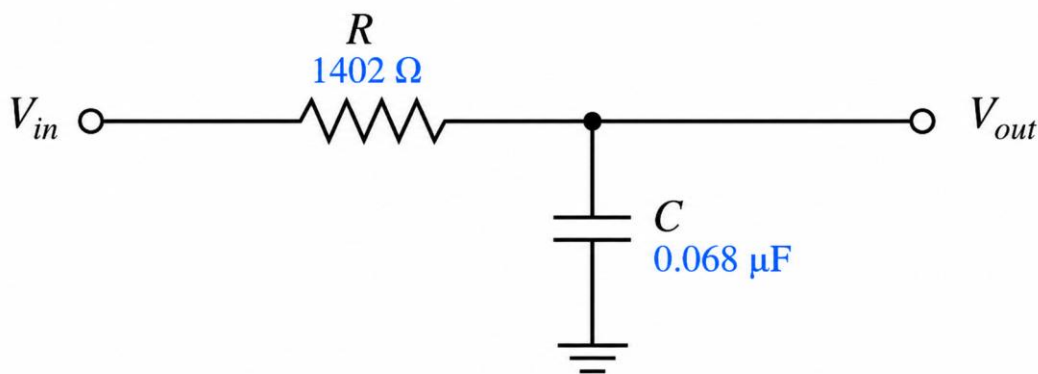
Aim: To study the frequency response of a first-order passive low-pass filter with target cutoff frequency of 1.7 KHz.

Apparatus:

Resistor, capacitor, function generator, oscilloscope, connecting wires and breadboard.

Theory:

A first-order passive low-pass filter can be constructed using a resistor R and capacitor C , as shown in Figure. The resistor is connected in series with the input, while the capacitor is connected to ground. The output voltage is taken across the capacitor.



First-Order RC Low-Pass Filter

The working of the circuit can be understood from the frequency dependence of the capacitive reactance,

$$X_C = \frac{1}{2\pi fC}.$$

At low frequencies, X_C is large. Hence, the capacitor offers a large impedance effectively acting as an open circuit and most of the input voltage appears across the capacitor. Therefore,

$$V_{out} \approx V_{in}.$$

As the frequency increases, the capacitive reactance decreases. A greater fraction of the input voltage is then dropped across the resistor, which causes the output voltage to decrease. Thus, the circuit passes low-frequency components while attenuating higher-frequency components.

The magnitude response can be obtained from the voltage-divider relation. The impedance of the capacitor is

$$Z_c = \frac{1}{j\omega C}$$

where $\omega = 2\pi f$. Therefore, the transfer function is

$$H(j\omega) = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{Z_c}{R + Z_c} = \frac{1}{1 + j\omega RC}.$$

Hence, the magnitude of the frequency response is

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

and the phase response is

$$\phi(\omega) = -\tan^{-1}(\omega RC)$$

The transfer function shows how the output voltage changes relative to the input voltage as the frequency is varied. From the magnitude response, at low frequencies, $\omega RC \ll 1$, so

$$|H(j\omega)| \approx 1.$$

Hence, the output voltage is nearly equal to the input voltage and the signal is passed with little attenuation. As the frequency increases, ωRC increases and the magnitude of the response decreases. Thus, higher-frequency components are progressively attenuated.

For the present first-order RC low-pass filter, using the conventional -3dB point as the cutoff frequency, we obtain

$$|H(j\omega_c)| = \frac{1}{\sqrt{2}}$$

Substituting the magnitude response gives

$$\frac{1}{\sqrt{1 + (\omega_c RC)^2}} = \frac{1}{\sqrt{2}},$$

which simplifies to

$$\omega_c RC = 1.$$

So, the cutoff angular frequency is obtained as

$$\omega_c = \frac{1}{RC}$$

or, in terms of frequency,

$$f_c = \frac{1}{2\pi RC}$$

At the cutoff frequency, the output voltage is expected to be approximately 70.7% of the input voltage with a roll-off rate of approximately -20dB/decade .

For the circuit used in the experiment,

$$R = 1402 \, \Omega, C = 0.068 \, \mu\text{F},$$

and hence

$$f_c = \frac{1}{2\pi(1402)(0.068 \times 10^{-6})} \approx 1.67 \times 10^3 \text{ Hz.}$$

Therefore,

$$f_c \approx 1.67 \text{ kHz}$$

Observations:

Components used: $R = 1402 \, \text{ohm}$; $C = 0.068 \, \mu\text{F}$

Target Cutoff Frequency (-3 dB point) = 1.67 KHz

Target Roll-off Rate = -20dB/decade .

Input Measured Voltage, V_{in} : 10.2 volts

Sr. No.	Frequency (Hz)	Output Voltage (V)	Gain (V_o/V_{in})	Gain(dB)= $20\log_{10}(V_o/V_{\text{in}})$
1	50	10.20	1.00	0.00
2	100	10.20	1.00	0.00
3	150	10.20	1.00	0.00
4	200	10.20	1.00	0.00

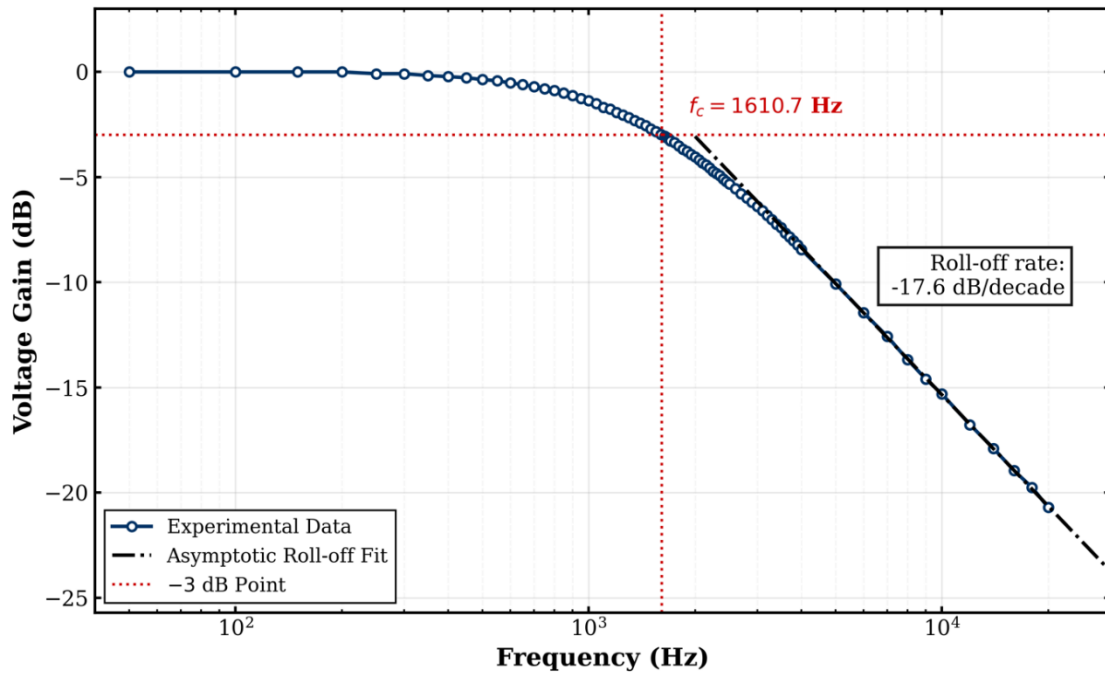
5	250	10.10	0.99	-0.09
6	300	10.10	0.99	-0.09
7	350	10.00	0.98	-0.17
8	400	9.94	0.97	-0.22
9	450	9.88	0.97	-0.28
10	500	9.78	0.96	-0.37
11	550	9.72	0.95	-0.42
12	600	9.60	0.94	-0.53
13	650	9.52	0.93	-0.60
14	700	9.41	0.92	-0.70
15	750	9.30	0.91	-0.80
16	800	9.21	0.90	-0.89
17	850	9.08	0.89	-1.01
18	900	8.96	0.88	-1.13
19	950	8.82	0.86	-1.26
20	1,000	8.71	0.85	-1.37
21	1,050	8.58	0.84	-1.50
22	1,100	8.41	0.82	-1.68
23	1,150	8.32	0.82	-1.77
24	1,200	8.17	0.80	-1.93
25	1,250	8.06	0.79	-2.05
26	1,300	7.94	0.78	-2.18
27	1,350	7.82	0.77	-2.31
28	1,400	7.71	0.76	-2.43
29	1,450	7.60	0.75	-2.56
30	1,500	7.47	0.73	-2.71
31	1,550	7.37	0.72	-2.82
32	1,600	7.24	0.71	-2.98
33	1,640	7.17	0.70	-3.06
34	1,650	7.15	0.70	-3.09
35	1,660	7.13	0.70	-3.11
36	1,670	7.11	0.70	-3.13
37	1,680	7.09	0.70	-3.16
38	1,690	7.05	0.69	-3.21
39	1,700	7.00	0.69	-3.27
40	1,750	6.94	0.68	-3.34
41	1,800	6.81	0.67	-3.51
42	1,850	6.68	0.65	-3.68
43	1,900	6.61	0.65	-3.77
44	1,950	6.50	0.64	-3.91
45	2,000	6.40	0.63	-4.05
46	2,050	6.31	0.62	-4.17
47	2,100	6.20	0.61	-4.32
48	2,150	6.13	0.60	-4.42

49	2,200	6.03	0.59	-4.57
50	2,250	5.92	0.58	-4.73
51	2,300	5.85	0.57	-4.83
52	2,350	5.79	0.57	-4.92
53	2,400	5.68	0.56	-5.09
54	2,450	5.60	0.55	-5.21
55	2,500	5.53	0.54	-5.32
56	2,600	5.38	0.53	-5.56
57	2,700	5.24	0.51	-5.79
58	2,800	5.12	0.50	-5.99
59	2,900	5.00	0.49	-6.19
60	3,000	4.89	0.48	-6.39
61	3,100	4.77	0.47	-6.60
62	3,200	4.65	0.46	-6.82
63	3,300	4.55	0.45	-7.01
64	3,400	4.43	0.43	-7.24
65	3,500	4.35	0.43	-7.40
66	3,600	4.22	0.41	-7.67
67	3,700	4.14	0.41	-7.83
68	3,800	4.05	0.40	-8.02
69	3,900	3.96	0.39	-8.22
70	4,000	3.86	0.38	-8.44
71	5,000	3.20	0.31	-10.07
72	6,000	2.73	0.27	-11.45
73	7,000	2.40	0.24	-12.57
74	8,000	2.11	0.21	-13.69
75	9,000	1.90	0.19	-14.60
76	10,000	1.75	0.17	-15.31
77	12,000	1.48	0.15	-16.77
78	14,000	1.30	0.13	-17.89
79	16,000	1.15	0.11	-18.96
80	18,000	1.05	0.10	-19.75
81	20,000	0.94	0.09	-20.71

Results and Discussions:

The magnitude response of the first-order passive low-pass filter is shown in Figure below.

First Order Passive Low Pass Filter



At low frequencies, the output voltage remains close to the input voltage of 10.2V, indicating that the low-frequency components are passed with very little attenuation. As the frequency increases, the output voltage decreases continuously, demonstrating the expected low-pass behaviour. The observation table also shows this gradual reduction in gain, with the gain decreasing from approximately 0dB at low frequencies to increasingly negative values at higher frequencies.

For the chosen values of R and C, the theoretical cutoff frequency was calculated to be $f_{c,\text{theory}} \approx 1.67 \text{ kHz}$. The experimental frequency response gives a -3dB frequency of approximately

$$f_{c,\text{exp}} \approx 1.61 \text{ kHz}.$$

Thus, the experimental value is close to the theoretical value, with a deviation of approximately 3.5%.

The experimental roll-off obtained from the asymptotic fit is approximately

$$-17.6 \text{ dB/decade}$$

which is reasonably close to the theoretical first-order value of -20 dB/decade . The difference is expected because the theoretical value represents the asymptotic slope of the ideal transfer function, whereas the experimental slope is obtained over a finite frequency range.

The small difference between the theoretical and experimental cutoff frequencies can arise from the actual component values, component tolerances, measurement uncertainty and others. Overall, the measured response agrees well with the theoretical first-order low-pass behaviour. The experimentally obtained cutoff frequency and roll-off confirm that the circuit behaves as a first-order passive low-pass filter.

Experiment 2: Second-Order Passive Low Pass Filter

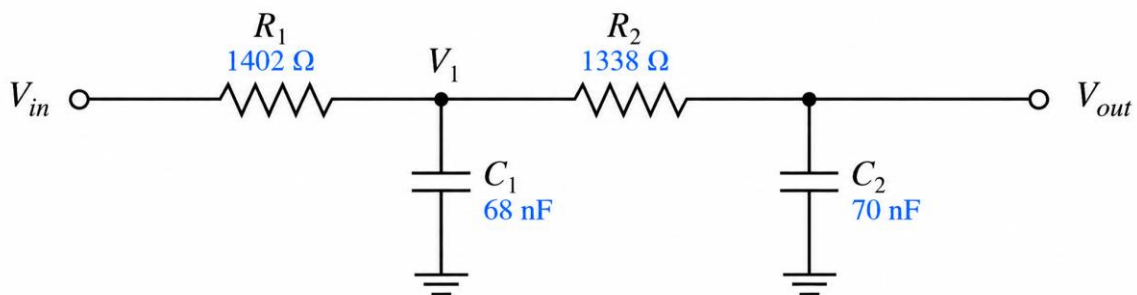
Aim: To study the frequency response of a second-order passive low-pass filter and determine its frequency-dependent attenuation and cutoff behaviour.

Apparatus:

Resistors, capacitors, function generator, oscilloscope, connecting wires and breadboard.

Theory:

A second-order passive low-pass filter can be constructed using two resistor-capacitor sections connected in cascade. The first section consists of R_1 and C_1 , while the second consists of R_2 and C_2 . The output voltage is taken across the second capacitor C_2 .



Second-Order RC Low-Pass Filter (Two Cascaded First-Order Stages)

For an idealised second-order low-pass filter, the transfer function can be expressed in the standard form

$$H(s) = \frac{\omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

where ω_0 is the characteristic angular frequency and Q is the quality factor. The characteristic frequency is determined by the product of the two RC time constants and is given by

$$\omega_0 = \frac{1}{\sqrt{R_1 C_1 R_2 C_2}}$$

or, in terms of frequency,

$$f_0 = \frac{1}{2\pi\sqrt{R_1 C_1 R_2 C_2}}.$$

This is called nominal or characteristic frequency. This should not be identified directly with the -3dB cutoff frequency for a higher order responses. To see the significance of this characteristic frequency, consider first the idealised case in which the two RC sections operate independently and have approximately the same characteristic frequency. The transfer function can then be written as

$$H_0(s) = \frac{1}{(1 + sR_1 C_1)(1 + sR_2 C_2)}.$$

For two sections having the same time constant,

$$R_1 C_1 = R_2 C_2 = RC,$$

this becomes

$$H_0(s) = \frac{1}{(1 + sRC)^2}.$$

At the characteristic angular frequency

$$\omega_0 = \frac{1}{RC}$$

the magnitude of each individual first-order section is

$$|H_1(j\omega_0)| = \frac{1}{\sqrt{1 + (\omega_0 RC)^2}} = \frac{1}{\sqrt{2}}.$$

Therefore, the magnitude of the combined response is

$$|H_0(j\omega_0)| = \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}.$$

Expressed in decibels,

$$20\log_{10}\left(\frac{1}{2}\right) = -6.02 \text{ dB}.$$

Hence, for two matched RC sections, the characteristic frequency corresponds to an attenuation of approximately -6.02dB in the **overall second-order response**. This occurs because each individual section contributes approximately -3.01dB at its characteristic frequency. The -6.02dB value therefore has a direct physical interpretation for the cascaded RC network.

However, it should not be regarded as the universal definition of the cutoff frequency of a second-order filter. The conventional -3dB point of the complete response is a separate frequency and is obtained by applying

$$|H_0(j\omega_{-3})| = \frac{1}{\sqrt{2}}.$$

For the identical-stage case,

$$\frac{1}{1 + (\omega_{-3}RC)^2} = \frac{1}{\sqrt{2}},$$

giving

$$\omega_{-3}RC = \sqrt{\sqrt{2} - 1},$$

and hence

$$\boxed{f_{-3} = \frac{\sqrt{\sqrt{2} - 1}}{2\pi RC} \approx 0.644 f_0.}$$

Thus, the characteristic frequency f_0 and the overall -3dB frequency are distinct for a second-order response.

For our case, the passive unbuffered circuit contains two energy-storage elements, C_1 and C_2 , and therefore has a second-order frequency response. However, the two passive sections in this circuit are directly connected without any buffer in between two stages. Hence, the second

section loads the first section, and the overall response will differ from the ideal second order response. This is regarded as *loading effect*.

Therefore, to obtain the actual response, we first calculate the transfer function of the circuit, let the voltage at the junction of R_1 , C_1 , and R_2 be V_1 , while the output voltage across C_2 is V_{out} . Applying Kirchhoff's current law at the two nodes gives

$$\frac{V_1 - V_{\text{in}}}{R_1} + sC_1V_1 + \frac{V_1 - V_{\text{out}}}{R_2} = 0$$

and

$$\frac{V_{\text{out}} - V_1}{R_2} + sC_2V_{\text{out}} = 0.$$

From the second equation,

$$V_1 = V_{\text{out}}(1 + sR_2C_2).$$

Substituting this relation into the first equation and simplifying gives the transfer function

$$H(s) = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{1}{1 + s(R_1C_1 + R_1C_2 + R_2C_2) + s^2R_1R_2C_1C_2}.$$

Thus, for $s = j\omega$,

$$H(j\omega) = \frac{1}{1 + j\omega(R_1C_1 + R_1C_2 + R_2C_2) - \omega^2R_1R_2C_1C_2}.$$

Defining

$$A = R_1C_1 + R_1C_2 + R_2C_2$$

and

$$B = R_1R_2C_1C_2,$$

the transfer function can be written as

$$H(j\omega) = \frac{1}{1 - \omega^2 B + j\omega A}.$$

Hence, the magnitude response is

$$|H(j\omega)| = \frac{1}{\sqrt{(1 - \omega^2 B)^2 + (\omega A)^2}}$$

At low frequencies, $\omega \rightarrow 0$, and therefore

$$|H(j\omega)| \rightarrow 1.$$

Thus, low-frequency components are passed with little attenuation. As the frequency increases, the denominator of the transfer function increases and the output voltage decreases. Since there are two reactive elements, the high-frequency attenuation is steeper than that of a first-order low-pass filter. The asymptotic roll-off for this second order circuit is expected to be -40 dB/decade or approximately -12 dB/octave.

For the design purpose, we use the formula of nominal or characteristic frequency or -6dB point for buffered response and realise the loading effect in the actual response. For the present circuit, we consider the following combinations,

$$\begin{aligned} R_1 &= 1402 \, \Omega, \, C_1 = 68 \, \text{nF}, \\ R_2 &= 1338 \, \Omega, \, C_2 = 70 \, \text{nF}. \end{aligned}$$

The individual section characteristic frequencies are

$$f_{c1} = \frac{1}{2\pi R_1 C_1} \approx 1.67 \, \text{kHz},$$

and

$$f_{c2} = \frac{1}{2\pi R_2 C_2} \approx 1.70 \, \text{kHz}.$$

Thus, the two sections are ideally expected to produce a second-order low-pass response with the characteristic frequency of each individual RC section lying close to 1.7 kHz.

However, to predict the actual response of the practical unbuffered circuit with loading effect, we use the actual transfer function. For the components used,

$$A = (1402)(68 \times 10^{-9}) + (1402)(70 \times 10^{-9}) + (1338)(70 \times 10^{-9})$$

$$A \approx 2.87136 \times 10^{-4} \text{ s},$$

and

$$B = (1402)(1338)(68 \times 10^{-9})(70 \times 10^{-9})$$

$$B \approx 8.9292 \times 10^{-9} \text{ s}^2.$$

The resulting transfer function is therefore

$$H(s) = \frac{1}{1 + (2.87136 \times 10^{-4})s + (8.9292 \times 10^{-9})s^2}.$$

For this unbuffered passive RC network, the -3dB cutoff frequency is obtained from

$$|H(j\omega_{-3\text{dB},l})| = \frac{1}{\sqrt{2}},$$

which gives approximately

$$f_{-3\text{dB},l} \approx 620 \text{ Hz}$$

This reduction of the actual cutoff frequency is a consequence of the loading between the two passive sections. So we now expect with this RC network, for input signal of frequency 620 Hz, the output voltage becomes approximately 70.7 percent and the asymptotic roll off rate to be close to -40dB/decade owing to the second order response of the circuit.

We can also do calculations for the expected actual -6dB frequency after loading effect by setting the magnitude of the transfer function to 0.5, which gives

$$f_{-6\text{dB},l} \approx 1056 \text{ Hz}$$

Observations:

Components used: $R_1 = 1402 \text{ } \Omega$; $C_1 = 68 \text{ nF}$ & $R_2 = 1338 \text{ } \Omega$; $C_2 = 70 \text{ nF}$

Nominal/Characteristic Frequency (-6 dB reference): 1.70 kHz

Expected Loaded -3 dB Frequency: 620 Hz

Expected Loaded -6 dB Frequency: 1056 Hz

Target Roll-off Rate = -40dB/decade

Input Measured Voltage, V_{in} : 10.2 volts

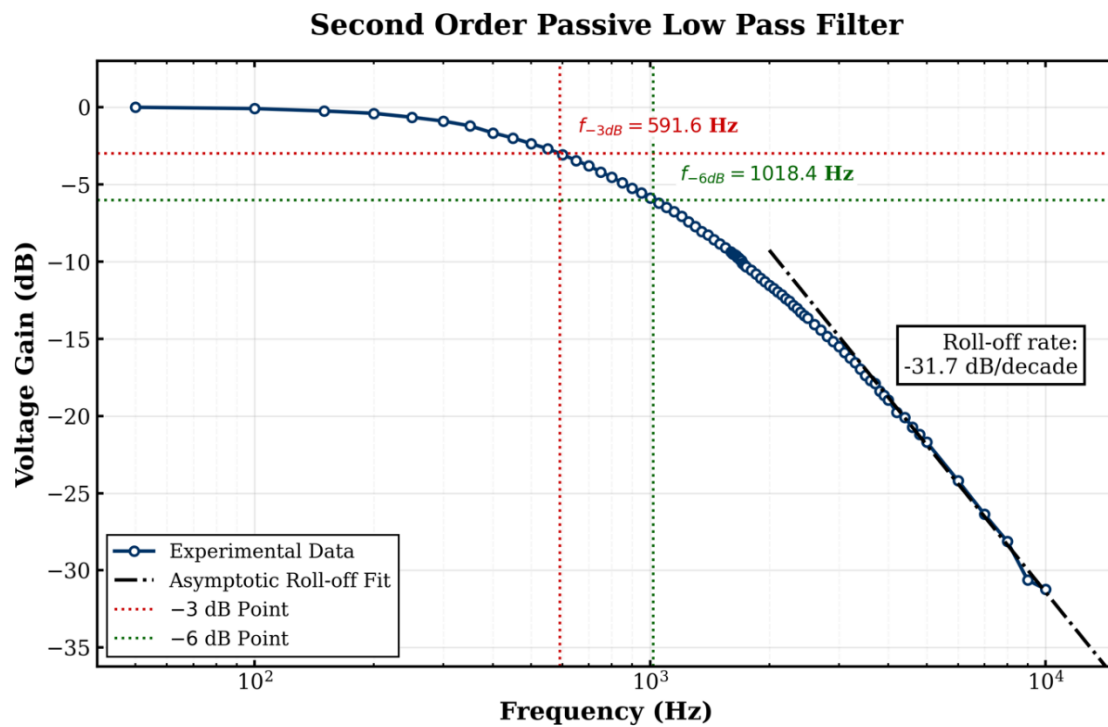
Sr. No.	Frequency (Hz)	Output Voltage (V)	Gain (V_o/V_{in})	Gain(dB)= $20\log_{10}(V_o/V_{in})$
1	50	10.20	1.00	0.00
2	100	10.10	0.99	-0.09
3	150	9.92	0.97	-0.24
4	200	9.75	0.96	-0.39
5	250	9.47	0.93	-0.65
6	300	9.20	0.90	-0.90
7	350	8.88	0.87	-1.20
8	400	8.43	0.83	-1.66
9	450	8.11	0.80	-1.99
10	500	7.78	0.76	-2.35
11	550	7.49	0.73	-2.68
12	600	7.17	0.70	-3.06
13	650	6.86	0.67	-3.45
14	700	6.59	0.65	-3.79
15	750	6.30	0.62	-4.19
16	800	6.05	0.59	-4.54
17	850	5.82	0.57	-4.87
18	900	5.57	0.55	-5.25
19	950	5.38	0.53	-5.56
20	1,000	5.18	0.51	-5.89
21	1,050	5.00	0.49	-6.19
22	1,100	4.84	0.47	-6.48
23	1,150	4.68	0.46	-6.77
24	1,200	4.52	0.44	-7.07
25	1,250	4.35	0.43	-7.40
26	1,300	4.20	0.41	-7.71
27	1,350	4.04	0.40	-8.04
28	1,400	3.94	0.39	-8.26
29	1,450	3.80	0.37	-8.58
30	1,500	3.68	0.36	-8.86
31	1,550	3.58	0.35	-9.09
32	1,600	3.47	0.34	-9.37
33	1,610	3.44	0.34	-9.44
34	1,620	3.41	0.33	-9.52
35	1,630	3.40	0.33	-9.54
36	1,640	3.39	0.33	-9.57
37	1,650	3.37	0.33	-9.62
38	1,660	3.33	0.33	-9.72

39	1,670	3.32	0.33	-9.75
40	1,680	3.30	0.32	-9.80
41	1,690	3.27	0.32	-9.88
42	1,700	3.26	0.32	-9.91
43	1,710	3.18	0.31	-10.12
44	1,720	3.17	0.31	-10.15
45	1,730	3.14	0.31	-10.23
46	1,740	3.13	0.31	-10.26
47	1,750	3.11	0.30	-10.32
48	1,800	3.03	0.30	-10.54
49	1,850	2.94	0.29	-10.81
50	1,900	2.85	0.28	-11.08
51	1,950	2.78	0.27	-11.29
52	2,000	2.70	0.26	-11.54
53	2,050	2.64	0.26	-11.74
54	2,100	2.58	0.25	-11.94
55	2,150	2.52	0.25	-12.14
56	2,200	2.45	0.24	-12.39
57	2,250	2.40	0.24	-12.57
58	2,300	2.33	0.23	-12.82
59	2,350	2.28	0.22	-13.01
60	2,400	2.21	0.22	-13.28
61	2,450	2.16	0.21	-13.48
62	2,500	2.12	0.21	-13.65
63	2,600	2.02	0.20	-14.06
64	2,700	1.94	0.19	-14.42
65	2,800	1.85	0.18	-14.83
66	2,900	1.78	0.17	-15.16
67	3,000	1.71	0.17	-15.51
68	3,100	1.64	0.16	-15.88
69	3,200	1.57	0.15	-16.25
70	3,300	1.52	0.15	-16.54
71	3,400	1.45	0.14	-16.94
72	3,500	1.38	0.14	-17.37
73	3,600	1.33	0.13	-17.69
74	3,700	1.30	0.13	-17.89
75	3,800	1.23	0.12	-18.37
76	3,900	1.19	0.12	-18.66
77	4,000	1.15	0.11	-18.96
78	4,200	1.05	0.10	-19.75
79	4,400	1.01	0.10	-20.09
80	4,600	0.94	0.09	-20.71

81	4,800	0.89	0.09	-21.18
82	5,000	0.84	0.08	-21.69
83	6,000	0.63	0.06	-24.19
84	7,000	0.49	0.05	-26.37
85	8,000	0.40	0.04	-28.13
86	9,000	0.30	0.03	-30.63
87	10,000	0.28	0.03	-31.23

Results and Discussions:

The magnitude response of the second-order passive low-pass filter is shown in Figure below.



The response remains close to 0dB at low frequencies and then decreases more rapidly with increasing frequency than the first-order filter. This confirms the stronger attenuation expected from a second-order low-pass response.

The actual circuit is an unbuffered passive RC cascade. The second RC section therefore loads the first section. The complete circuit transfer function derived in the theory section predicts a -3dB frequency of approximately $f_{-3,\text{theory}} \approx 620 \text{ Hz}$ and a -6dB frequency of approximately $f_{-6,\text{theory}} \approx 1056 \text{ Hz}$.

The experimental plot gives

$$f_{-3,\text{exp}} \approx 591.6 \text{ Hz}$$

and

$$f_{-6,\text{exp}} \approx 1018.4 \text{ Hz}.$$

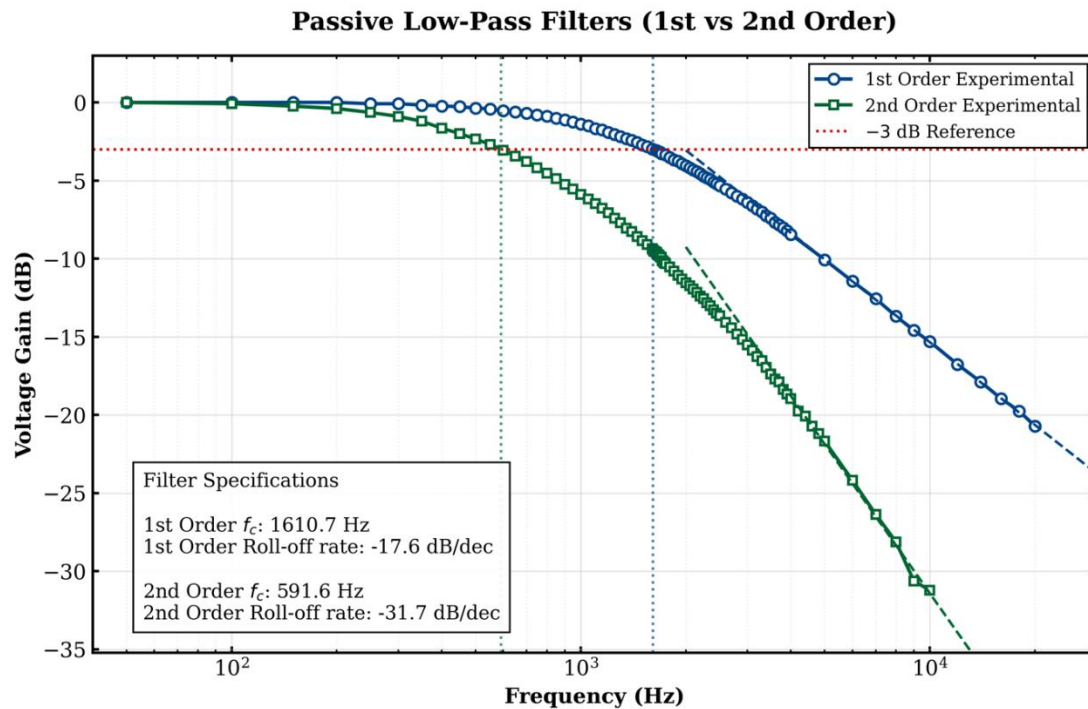
The experimental values are therefore close to the values predicted from the complete loaded transfer function, with deviations of approximately 4.6% for the -3dB point and 3.6% for the -6dB point.

A significant difference can be observed between the nominal characteristic frequency of approximately 1.70kHz and the experimentally obtained -6dB frequency of approximately 1.02kHz. This shift demonstrates the effect of loading between the two passive sections. The nominal frequency is therefore useful as a design reference for the two RC sections, but it does not represent the actual attenuation frequency of the complete unbuffered circuit.

The experimental roll-off obtained from the asymptotic fit is approximately

$$-31.7 \text{ dB/decade}$$

whereas the ideal second-order asymptotic value is -40 dB/decade . The measured value is therefore less steep than the ideal asymptotic value. The -40dB/decade value describes the asymptotic high-frequency behaviour of an ideal second-order response, while the experimental fit is obtained over a finite frequency range in which the response may not yet have reached its full asymptotic slope.



The comparison plot of the first- and second-order experimental responses shown above further illustrates the behaviour of first and second order circuits. The second-order filter begins to attenuate at a lower frequency and its gain decreases more rapidly with increasing frequency. At higher frequencies, the second-order response therefore provides substantially greater attenuation than the first-order filter. This is consistent with the additional reactive element and the resulting second-order transfer function.

The difference between the two experimental responses also demonstrates the practical importance of loading in passive filter design. If the two RC sections were isolated by a buffer, the response would more closely follow the product of two independent first-order responses. In the present circuit, direct connection allows the second section to draw current from the first, which modifies the effective response of the first section and shifts the characteristic attenuation frequencies.

Overall, the experimental second-order response agrees with the qualitative behaviour predicted by the theoretical model. The measured -3dB and -6dB frequencies are close to those obtained from the complete loaded transfer function, while the deviation from the nominal characteristic frequency demonstrates the loading effect. The comparison with the first-order filter also confirms the greater attenuation provided by the second-order response.

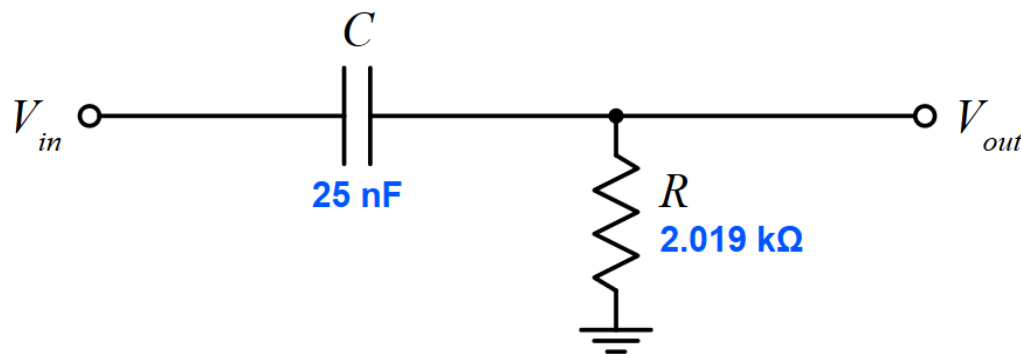
Experiment 3: First Order Passive High Pass Filter (HPF)-

Aim: To study the frequency response of a first-order passive high-pass filter with target cutoff frequency of approximately 3.2 kHz.

Apparatus:

Resistor, capacitor, function generator, oscilloscope, connecting wires and breadboard.

Circuit Diagram:



First-Order RC High-Pass Filter

Theory:

A first-order passive high-pass filter can be constructed using a series capacitor C and a resistor R connected to ground, as shown in Figure 1. The output voltage is taken across the resistor. The capacitor blocks low-frequency components more strongly, while allowing higher-frequency components to appear at the output.

The working of the circuit can be understood from the frequency dependence of the capacitive reactance,

$$X_C = \frac{1}{2\pi fC}$$

At low frequencies, the capacitive reactance is large. Hence, the capacitor offers a large impedance and only a small fraction of the input voltage appears across the resistor. As the frequency increases, the capacitive reactance decreases and a greater fraction of the input voltage appears across the resistor. Therefore, the circuit passes high-frequency components while attenuating low-frequency components.

The impedance of the capacitor is given by-

$$Z_C = \frac{1}{j\omega C}$$

where $\omega = 2\pi f$. Since the output is taken across R, the voltage-divider relation gives the transfer function

$$H(j\omega) = \frac{V_{out}}{V_{in}} = \frac{R}{R + Z_C} = \frac{j\omega RC}{1 + j\omega RC}$$

Hence, the magnitude response is

$$|H(j\omega)| = \frac{\omega RC}{\sqrt{1 + (\omega RC)^2}}$$

and the phase response is

$$\phi(\omega) = 90^\circ - \tan^{-1}(\omega RC)$$

For the conventional -3 dB cutoff point, the magnitude of the response is $1/\sqrt{2}$ of its high-frequency value. This gives

$$\omega_c RC = 1 \quad \Rightarrow \quad f_c = \frac{1}{2\pi RC}$$

At the cutoff frequency, the output voltage is approximately 70.7% of the input voltage and the first-order roll-off rate beyond the transition region is approximately +20 dB/decade.

For the circuit used in the experiment,

$C = 25 \text{ nF}$, $R(\text{design}) \approx 1.989 \text{ k}\Omega$, $R(\text{experimental}) \approx 2.019 \text{ k}\Omega$.

Using the experimental resistance,

$$f_c = \frac{1}{2\pi(2019)(25 \times 10^{-9})} \approx 3.15 \times 10^3 \text{ Hz}$$

Therefore, the calculated cutoff frequency for the experimental component values is approximately 3.15 kHz.

Observations:

Components used: $R \approx 2.019 \text{ k}\Omega$, $C = 25 \text{ nF}$

Target Cutoff Frequency (-3 dB point) $\approx 3.2 \text{ kHz}$

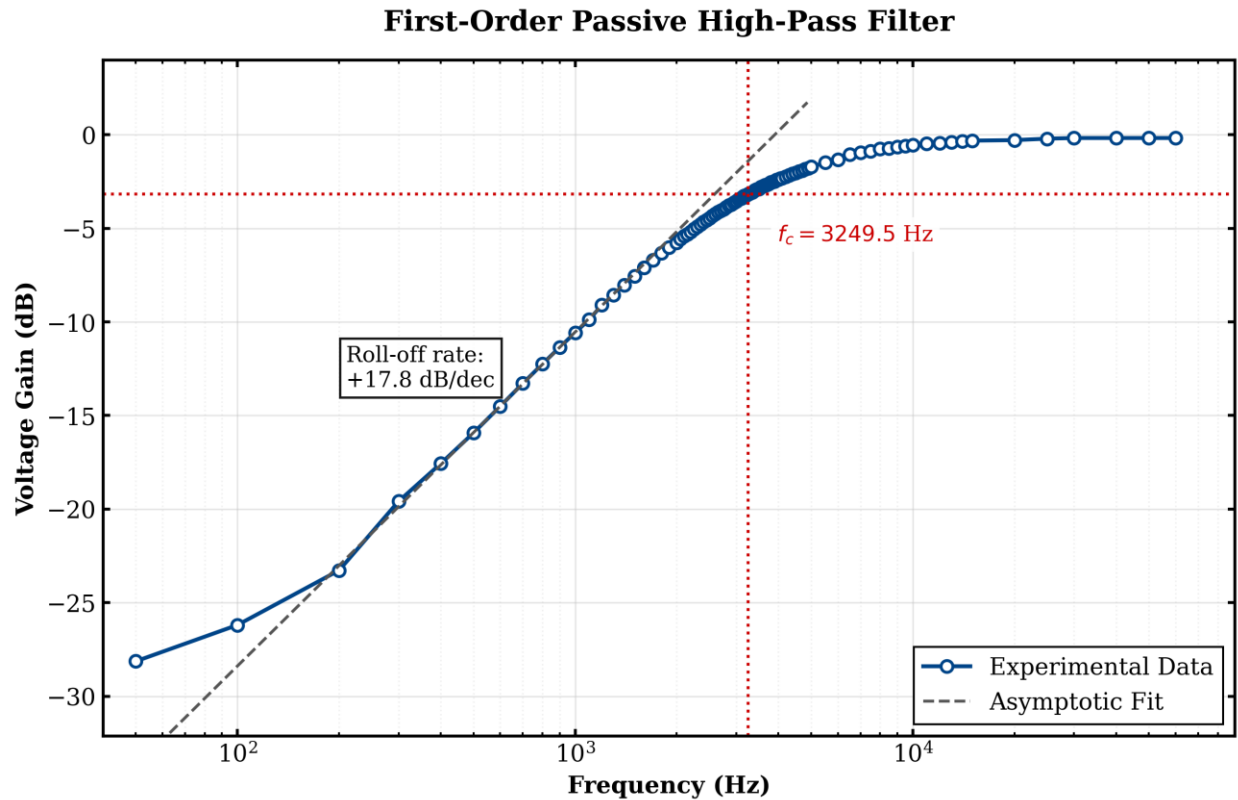
Input Measured Voltage, $V_{in} \approx 10 \text{ V}$

Sr. No.	Frequency (Hz)	Output Voltage (V)	Gain (V_o/V_{in})	Gain(dB)= $20\log_{10}(V_o/V_{in})$
1	50	0.4	0.040	-27.96
2	100	0.5	0.050	-26.02
3	200	0.7	0.070	-23.10
4	300	1.07	0.107	-19.41
5	400	1.35	0.135	-17.39
6	500	1.63	0.163	-15.75
7	600	1.92	0.192	-14.33
8	700	2.21	0.221	-13.11
9	800	2.49	0.249	-12.08
10	900	2.76	0.276	-11.18
11	1000	3.02	0.302	-10.40
12	1100	3.27	0.327	-9.71
13	1200	3.58	0.358	-8.92
14	1300	3.81	0.381	-8.38
15	1400	4.05	0.405	-7.85
16	1500	4.28	0.428	-7.37
17	1600	4.5	0.450	-6.94
18	1700	4.72	0.472	-6.52
19	1800	4.93	0.493	-6.14
20	1900	5.1	0.510	-5.85
21	2000	5.26	0.526	-5.58
22	2050	5.38	0.538	-5.38
23	2100	5.47	0.547	-5.24
24	2150	5.53	0.553	-5.15
25	2200	5.61	0.561	-5.02
26	2250	5.71	0.571	-4.87
27	2300	5.8	0.580	-4.73
28	2350	5.88	0.588	-4.61
29	2400	5.96	0.596	-4.51
30	2450	6.04	0.604	-4.38
31	2500	6.11	0.611	-4.18
32	2550	6.19	0.619	-4.17
33	2600	6.27	0.627	-4.05
34	2650	6.34	0.634	-3.96

Sr. No.	Frequency (Hz)	Output Voltage (V)	Gain (V_o/V_{in})	Gain(dB)= $20\log_{10}(V_o/V_{in})$
35	2700	6.4	0.640	-3.88
36	2750	6.46	0.646	-3.80
37	2800	6.53	0.653	-3.70
38	2850	6.6	0.660	-3.61
39	2900	6.65	0.665	-3.54
40	2950	6.73	0.673	-3.44
41	3000	6.78	0.678	-3.38
42	3050	6.86	0.686	-3.27
43	3100	6.91	0.691	-3.21
44	3110	6.92	0.692	-3.20
45	3120	6.93	0.693	-3.19
46	3130	6.94	0.694	-3.17
47	3140	6.95	0.695	-3.16
48	3150	6.97	0.697	-3.14
49	3160	6.97	0.697	-3.14
50	3170	7	0.700	-3.10
51	3180	7.01	0.701	-3.09
52	3190	7.02	0.702	-3.07
53	3200	7.02	0.702	-3.07
54	3250	7.08	0.708	-3.00
55	3300	7.13	0.713	-2.94
56	3350	7.17	0.717	-2.89
57	3400	7.24	0.724	-2.81
58	3450	7.3	0.730	-2.73
59	3500	7.32	0.732	-2.71
60	3550	7.39	0.739	-2.63
61	3600	7.44	0.744	-2.57
62	3650	7.48	0.748	-2.52
63	3700	7.51	0.751	-2.49
64	3750	7.56	0.756	-2.43
65	3800	7.61	0.761	-2.37
66	3850	7.64	0.764	-2.34
67	3900	7.67	0.767	-2.30
68	3950	7.72	0.772	-2.25

Sr. No.	Frequency (Hz)	Output Voltage (V)	Gain (V_o/V_{in})	Gain(dB)= $20\log_{10}(V_o/V_{in})$
69	4000	7.76	0.776	-2.20
70	4050	7.79	0.779	-2.17
71	4100	7.81	0.781	-2.15
72	4200	7.89	0.789	-2.06
73	4300	7.96	0.796	-1.98
74	4400	8.02	0.802	-1.92
75	4500	8.06	0.806	-1.87
76	4600	8.15	0.815	-1.78
77	4700	8.2	0.820	-1.72
78	4800	8.25	0.825	-1.67
79	4900	8.3	0.830	-1.62
80	5000	8.37	0.837	-1.55
81	5500	8.6	0.860	-1.31
82	6000	8.76	0.876	-1.15
83	6500	9.04	0.904	-0.88
84	7000	9.14	0.914	-0.78
85	7500	9.22	0.922	-0.71
86	8000	9.36	0.936	-0.57
87	8500	9.38	0.938	-0.56
88	9000	9.46	0.946	-0.48
89	9500	9.53	0.953	-0.42
90	10000	9.58	0.958	-0.37
91	11000	9.67	0.967	-0.29
92	12000	9.69	0.969	-0.27
93	13000	9.74	0.974	-0.23
94	14000	9.81	0.981	-0.17
95	15000	9.84	0.984	-0.14
96	20000	9.87	0.987	-0.11
97	25000	9.96	0.996	-1.12
98	30000	10	1	0
99	40000	10	1	0
100	50000	10	1	0
101	60000	10	1	0

Frequency Response:



Result and Discussion:

The frequency response of the **first-order passive high-pass filter** was obtained experimentally. At low frequencies, the gain is strongly attenuated, and it increases gradually with frequency. The experimental cutoff frequency was found to be approximately **3249.5 Hz**, at which the gain is about **-3 dB**. At higher frequencies, the gain approaches nearly **0 dB**, indicating that the high-frequency components are passed with little attenuation.

The measured roll-off rate is approximately **+17.8 dB/decade**, which is reasonably close to the theoretical value of **+20 dB/decade** for a first-order high-pass filter.

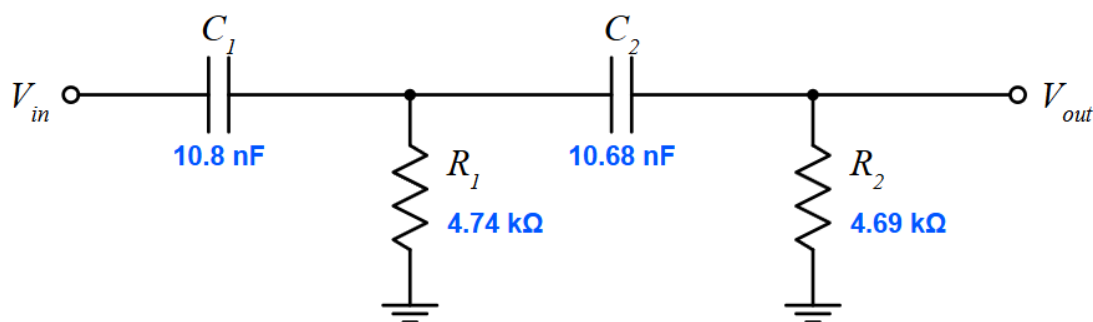
Experiment 4: Second Order Passive High Pass Filter-

Aim: To study the frequency response of a second-order passive high-pass filter formed by cascading two RC high-pass sections for cut off frequency of 3.2kHz.

Apparatus:

Resistors, capacitors, function generator, oscilloscope, connecting wires and breadboard.

Circuit Diagram:



Second-Order RC High-Pass Filter

Theory:

A second-order passive high-pass response can be obtained by cascading two first-order RC high-pass sections. The circuit used in the experiment consists of two series capacitors with resistors connected to ground, as shown in Figure 3. The output is taken across the second resistor.

Transfer Function of Second-Order High-Pass Filter

For an ideal buffered cascade, the two first-order high-pass sections can be treated independently. The transfer function of an individual RC high-pass section is

$$H_i(s) = \frac{sR_iC_i}{1 + sR_iC_i}$$

Therefore, for two cascaded sections, the idealized overall transfer function is-

$$H_0(s) = \frac{s^2R_1R_2C_1C_2}{(1 + sR_1C_1)(1 + sR_2C_2)}$$

The characteristic frequency of the two-stage network is determined by the product of the two RC time constants:

$$\omega_0 = \frac{1}{\sqrt{R_1 C_1 R_2 C_2}}$$

or, in terms of frequency,

$$f_0 = \frac{1}{2\pi\sqrt{R_1 C_1 R_2 C_2}}$$

Identical-Stage Case

For two sections having the same time constant,

$$R_1 C_1 = R_2 C_2 = RC$$

the transfer function becomes

$$H_0(s) = \left(\frac{sRC}{1 + sRC} \right)^2$$

At the characteristic angular frequency $\omega_0 = 1/(RC)$, the magnitude of each individual first-order section is $1/\sqrt{2}$. Hence, the magnitude of the combined second-order response is

$$|H_0(j\omega_0)| = \left(\frac{1}{\sqrt{2}} \right)^2 = \frac{1}{2}$$

Expressed in decibels, this corresponds to approximately -6.02 dB:

$$20\log_{10}\left(\frac{1}{2}\right) = -6.02 \text{ dB}$$

Thus, for two matched RC sections, the characteristic frequency corresponds to an attenuation of approximately -6.02 dB in the idealized overall response. This is different from the conventional -3 dB cutoff frequency of the complete second-order response.

Conventional -3 dB Frequency for the idealized Cascade

For the conventional -3 dB point of the complete response,

$$|H_0(j\omega_{3\text{dB}})| = \frac{1}{\sqrt{2}}$$

For identical stages, the magnitude response is

$$|H_0(j\omega)| = \frac{(\omega RC)^2}{1 + (\omega RC)^2}$$

Substituting the -3 dB condition gives

$$\frac{(\omega_{3dB}RC)^2}{1 + (\omega_{3dB}RC)^2} = \frac{1}{\sqrt{2}}$$

Solving for the frequency ratio gives

$$\omega_{3dB}RC = \sqrt{\sqrt{2} + 1} \approx 1.554$$

Therefore,

$$f_{3dB} \approx 1.554 f_0$$

Hence, the characteristic frequency and the overall -3 dB frequency are distinct for a second-order high-pass response.

Actual Passive Unbuffered Circuit and Loading Effect

The circuit used in the experiment is passive and unbuffered. Therefore, the second RC section is directly connected to the first section and loads it. To obtain the actual response, the transfer function of the complete circuit is derived using Kirchhoff's current law.

Let V_1 be the voltage at the junction of C_1 , R_1 and C_2 , and let V_{out} be the voltage across R_2 . Applying Kirchhoff's current law at the two nodes gives

$$sC_1(V_1 - V_{in}) + \frac{V_1}{R_1} + sC_2(V_1 - V_{out}) = 0$$

$$sC_2(V_{out} - V_1) + \frac{V_{out}}{R_2} = 0$$

From the second equation,

$$V_1 = V_{out}(1 + sR_2C_2)$$

Substituting this relation into the first node equation and simplifying gives the transfer function of the actual unbuffered passive network:

$$H(s) = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{s^2 R_1 R_2 C_1 C_2}{1 + s(R_1 C_1 + R_1 C_2 + R_2 C_2) + s^2 R_1 R_2 C_1 C_2}$$

Frequency Response of the Unbuffered Network

For sinusoidal steady-state operation, putting $s = j\omega$ gives

$$H(j\omega) = \frac{-\omega^2 B}{1 - \omega^2 B + j\omega A}$$

where,

$$A = R_1 C_1 + R_1 C_2 + R_2 C_2$$

$$B = R_1 R_2 C_1 C_2$$

Hence, the magnitude response is

$$|H(j\omega)| = \frac{\omega^2 B}{\sqrt{(1 - \omega^2 B)^2 + (\omega A)^2}}$$

At very low frequencies, $\omega \rightarrow 0$, so $|H(j\omega)| \rightarrow 0$. Thus, low-frequency components are strongly attenuated. At high frequencies, the ω^2 terms dominate and the magnitude approaches unity. Therefore, the circuit has the expected high-pass behaviour. Since the numerator contains s^2 , the asymptotic roll-up rate is approximately +40 dB/decade.

Calculation for the Components Used in the Experiment

For the experimental circuit, the component values are

$$R_1 = 4.74 \text{ k}\Omega, \quad R_2 = 4.69 \text{ k}\Omega, \quad C_1 = 10.8 \text{ nF}, \quad C_2 = 10.68 \text{ nF}$$

The individual RC characteristic frequencies are approximately

$$f_1 = \frac{1}{2\pi R_1 C_1} \approx 3.14 \text{ kHz}$$

$$f_2 = \frac{1}{2\pi R_2 C_2} \approx 3.14 \text{ kHz}$$

These values are close to the nominal target of 3.2 kHz. For the actual unbuffered network,

$$A \approx 1.524672 \times 10^{-4} \text{ s}$$

$$B \approx 2.564166 \times 10^{-9} \text{ s}^2$$

Therefore, the actual transfer function becomes

$$H(s) = \frac{(2.564166 \times 10^{-9})s^2}{1 + (1.524672 \times 10^{-4})s + (2.564166 \times 10^{-9})s^2}$$

For the conventional -3 dB condition, $|H(j\omega)| = 1/\sqrt{2}$. Using the above transfer function gives an expected -3 dB frequency of approximately

$$f_{3\text{dB,actual}} \approx 8.44 \text{ kHz}$$

The experimental observation gives an overall -3 dB frequency of approximately 8.84 kHz. The small difference between the calculated and measured values can arise from component tolerances, measurement uncertainty and practical loading in the experimental setup.

Observations:

Components used: $R1 \approx 4.74 \text{ k}\Omega$, $R2 \approx 4.69 \text{ k}\Omega$, $C1 \approx 10.8 \text{ nF}$, $C2 \approx 10.68 \text{ nF}$

Nominal stage cutoff frequency $\approx 3.2 \text{ kHz}$

Input Measured Voltage, $V_{in} \approx 10 \text{ V}$

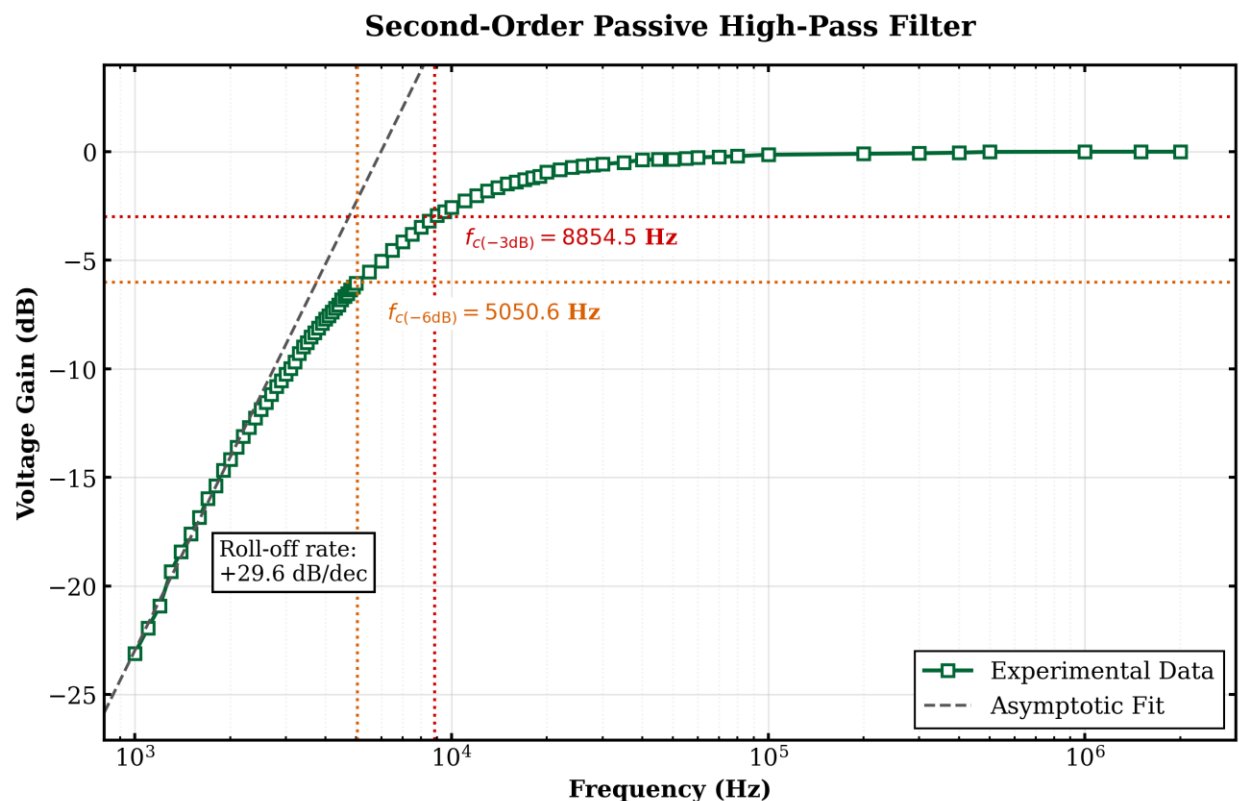
Sr. No.	Frequency (Hz)	Output Voltage (V)	Gain (V_o/V_{in})	Gain(dB)= $20\log_{10}(V_o/V_{in})$
1	1000	0.7	0.070	-23.10
2	1100	0.8	0.080	-21.94
3	1200	0.9	0.090	-20.92
4	1300	1.08	0.108	-19.33
5	1400	1.2	0.12	-18.41
6	1500	1.32	0.132	-17.59
7	1600	1.44	0.144	-16.83
8	1700	1.59	0.159	-15.97

Sr. No.	Frequency (Hz)	Output Voltage (V)	Gain (V_o/V_{in})	Gain(dB)= $20\log_{10}(V_o/V_{in})$
9	1800	1.7	0.170	-15.39
10	1900	1.85	0.185	-14.66
11	2000	1.96	0.196	-14.15
12	2100	2.09	0.209	-13.60
13	2200	2.21	0.221	-13.11
14	2300	2.32	0.232	-12.69
15	2400	2.44	0.244	-12.25
16	2500	2.55	0.255	-11.87
17	2600	2.65	0.265	-11.54
18	2700	2.76	0.276	-11.18
19	2800	2.88	0.288	-10.81
20	2900	2.97	0.297	-10.54
21	3000	3.08	0.308	-10.23
22	3100	3.17	0.317	-9.98
23	3200	3.28	0.328	-9.68
24	3300	3.43	0.343	-9.29
25	3400	3.56	0.356	-8.97
26	3500	3.64	0.364	-8.78
27	3600	3.75	0.375	-8.52
28	3700	3.83	0.383	-8.34
29	3800	3.92	0.392	-8.13
30	3900	4.03	0.403	-7.89
31	4000	4.12	0.412	-7.70
32	4100	4.19	0.419	-7.56
33	4200	4.28	0.428	-7.37
34	4300	4.36	0.436	-7.21
35	4400	4.44	0.444	-7.05
36	4500	4.56	0.456	-6.82
37	4600	4.64	0.464	-6.67
38	4700	4.72	0.472	-6.52
39	4800	4.81	0.481	-6.36
40	4900	4.88	0.488	-6.23
41	5000	4.98	0.498	-6.06
42	5500	5.29	0.529	-5.53

Sr. No.	Frequency (Hz)	Output Voltage (V)	Gain (V_o/V_{in})	Gain(dB)= $20\log_{10}(V_o/V_{in})$
43	6000	5.6	0.560	-5.04
44	6500	5.93	0.593	-4.54
45	7000	6.21	0.621	-4.14
46	7500	6.46	0.646	-3.80
47	8000	6.71	0.671	-3.47
48	8500	6.93	0.693	-3.19
49	9000	7.14	0.714	-2.93
50	9500	7.28	0.728	-2.76
51	10000	7.44	0.744	-2.57
52	11000	7.71	0.771	-2.26
53	12000	7.93	0.793	-2.01
54	13000	8.12	0.812	-1.81
55	14000	8.28	0.828	-1.64
56	15000	8.43	0.843	-1.48
57	16000	8.52	0.852	-1.39
58	17000	8.62	0.862	-1.29
59	18000	8.71	0.871	-1.20
60	19000	8.79	0.879	-1.12
61	20000	8.98	0.898	-0.93
62	22000	9.10	0.910	-0.81
63	24000	9.2	0.920	-0.72
64	26000	9.28	0.928	-0.65
65	28000	9.33	0.933	-0.60
66	30000	9.36	0.936	-0.57
67	35000	9.45	0.945	-0.49
68	40000	9.58	0.958	-0.37
69	45000	9.61	0.961	-0.35
70	50000	9.62	0.962	-0.34
71	55000	9.66	0.966	-0.30
72	60000	9.70	0.970	-0.26
73	70000	9.74	0.974	-0.23
74	80000	9.77	0.977	-0.20
75	100000	9.77	0.977	-0.20
76	200000	9.84	0.984	-0.14

Sr. No.	Frequency (Hz)	Output Voltage (V)	Gain (V_o/V_{in})	Gain(dB)= $20\log_{10}(V_o/V_{in})$
77	300000	9.89	0.989	-0.10
78	400000	9.92	0.992	-0.07
79	500000	9.95	0.995	-0.04
80	1000000	9.99	0.999	-0.01
81	1500000	10	1.000	0.00
82	2000000	10.1	1.010	0.09

Frequency Response:

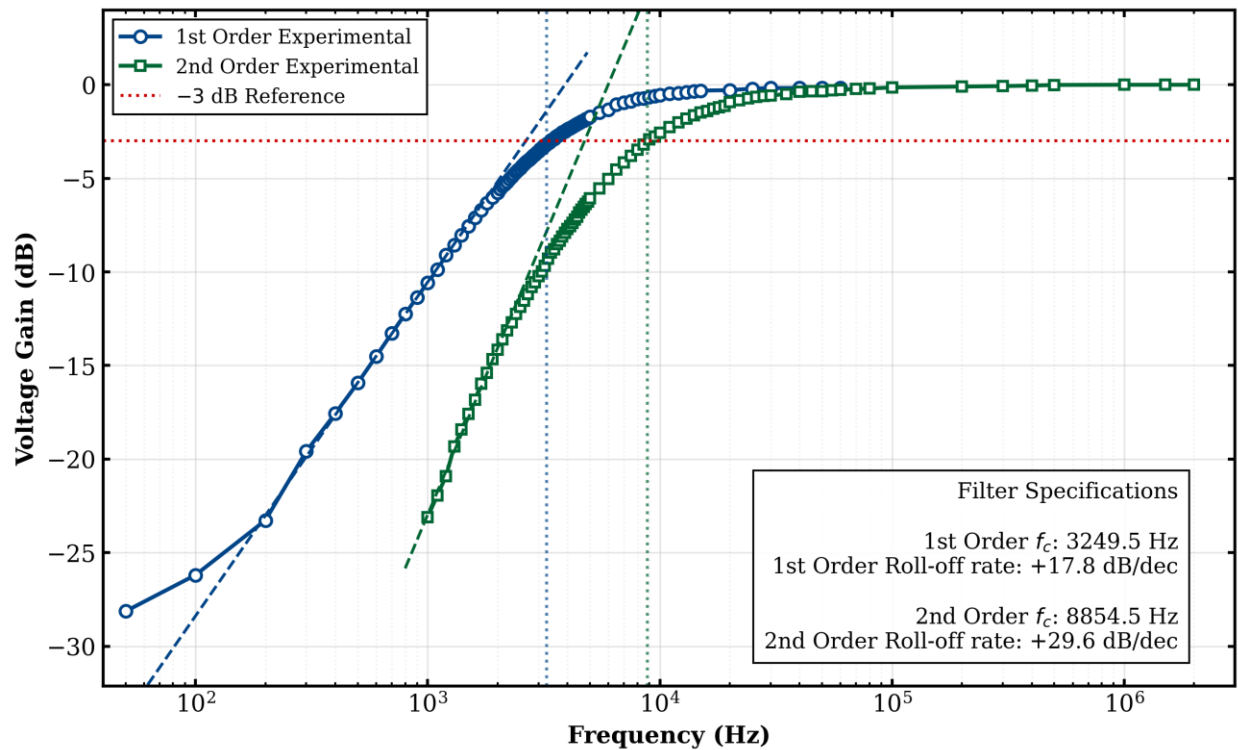


Result and Discussion:

The frequency response of the second-order passive high-pass filter was obtained experimentally. The gain increases with frequency and gradually approaches 0 dB in the passband, showing the expected high-pass characteristics. The measured cutoff frequencies were 5050.6 Hz at -6 dB and 8854.5 Hz at -3 dB. The experimental roll-off rate was approximately +29.6 dB/decade, which is lower than the theoretical +40 dB/decade expected for a second-order high-pass filter.

Comparison of 1st order and 2nd order High Pass Filter:

Passive High-Pass Filters (1st vs 2nd Order)



This graph compares the frequency responses of the first-order and second-order passive high-pass filters. Both filters exhibit the expected high-pass characteristics, with strong attenuation at low frequencies and nearly constant gain at higher frequencies. The first-order filter has an experimental cutoff frequency of 3249.5 Hz and a roll-off rate of +17.8 dB/decade. In comparison, the second-order filter shows a cutoff frequency of 8854.5 Hz and a roll-off rate of +29.6 dB/decade.

The second-order filter provides a sharper transition from the stopband to the passband than the first-order filter, as indicated by its higher roll-off rate. Ideally, the roll-off rate should be approximately +20 dB/decade for a first-order filter and +40 dB/decade for a second-order filter. The experimental values are lower than the theoretical values, particularly for the second-order filter. This deviation may be due to component tolerances, loading effects between the filter sections, parasitic effects, and practical limitations of the components. Overall, the comparison clearly demonstrates that increasing the filter order improves the steepness of the frequency response.

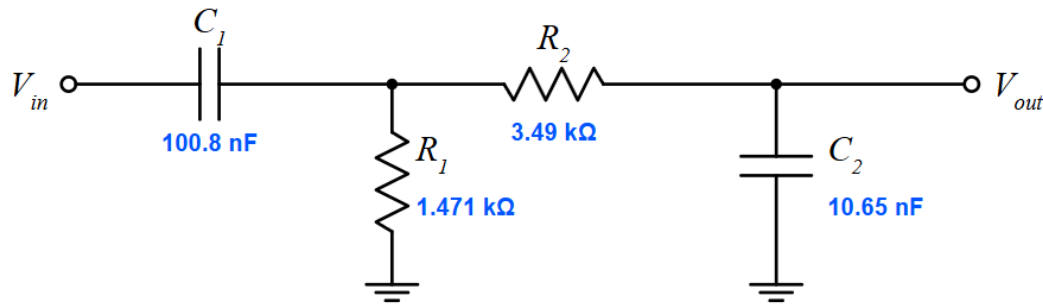
Experiment 5: Passive Band Pass Filter

Aim: To study the frequency response of a passive band-pass filter for cut off frequencies between 1kHz to 4kHz.

Apparatus:

Resistors, capacitors, function generator, oscilloscope, connecting wires and breadboard.

Circuit Diagram:



Passive RC Band-Pass Filter

Theory:

A passive band-pass filter allows a selected range of frequencies to pass while attenuating frequencies below the lower cutoff and above the upper cutoff. A practical passive band-pass response can be obtained by combining a high-pass RC section for the lower-frequency boundary and a low-pass RC section for the upper-frequency boundary.

In the present filter, the high-pass section uses R_1 and C_1 , while the low-pass section uses R_2 and C_2 . At low frequencies, the high-pass capacitor has high reactance and strongly attenuates the output. As the frequency increases, the high-pass section passes a larger fraction of the signal. At sufficiently high frequencies, the low-pass capacitor reduces the output, so the response again decreases. Therefore, the response is maximum in the intermediate frequency range.

Transfer Functions of the Individual Sections

For the idealized first-order high-pass section, the transfer function is-

$$H_{HP}(s) = \frac{sR_1C_1}{1 + sR_1C_1}$$

For the idealised first-order low-pass section, the transfer function is

$$H_{LP}(s) = \frac{1}{1 + sR_2C_2}$$

For an idealised cascade in which loading between the two sections is neglected, the overall band-pass transfer function is the product of the two section responses:

$$H_{BPF}(s) = \frac{sR_1C_1}{(1 + sR_1C_1)(1 + sR_2C_2)}$$

The practical passive circuit does not contain an active buffer between the two sections. Therefore, loading between the high-pass and low-pass networks can shift the measured response from the idealised values. The experimental cutoff frequencies are therefore obtained directly from the measured frequency-response data.

Cutoff Frequencies

For the high-pass RC section, the nominal lower cutoff frequency is

$$f_L = \frac{1}{2\pi R_1 C_1}$$

For the low-pass RC section, the nominal upper cutoff frequency is

$$f_H = \frac{1}{2\pi R_2 C_2}$$

Component Values and Nominal Frequencies

For the high-pass section, the component values are $R_1 = 1.471 \text{ k}\Omega$ and $C_1 = 100.8 \text{ nF}$. Therefore,

$$f_L = \frac{1}{2\pi(1.471 \times 10^3)(100.8 \times 10^{-9})} \approx 1.073 \text{ kHz}$$

For the low-pass section, the component values are $R_2 = 3.49 \text{ k}\Omega$ and $C_2 = 10.65 \text{ nF}$. Therefore,

$$f_H = \frac{1}{2\pi(3.49 \times 10^3)(10.65 \times 10^{-9})} \approx 4.282 \text{ kHz}$$

Centre Frequency, Bandwidth and Quality Factor

For a band-pass response, the nominal centre frequency is approximated by the geometric mean of the lower and upper cutoff frequencies:

$$f_c = \sqrt{f_L f_H} = \sqrt{(1073)(4282)} \approx 2.144 \text{ kHz}$$

The nominal bandwidth is the difference between the upper and lower cutoff frequencies:

$$B = f_H - f_L = 4.282 - 1.073 \approx 3.209 \text{ kHz}$$

The corresponding nominal quality factor is

$$Q = \frac{f_c}{B} \approx \frac{2.144}{3.209} \approx 0.668$$

Experimental Cutoff Frequencies

From the observation table, the maximum measured output voltage is 7.28 V at 1.9 kHz. Taking this as the measured peak response, the -3 dB output level relative to the peak is

$$V_{-3 \text{ dB}} = \frac{V_{\max}}{\sqrt{2}} = \frac{7.28}{\sqrt{2}} \approx 5.15 \text{ V}$$

The lower -3 dB frequency is between 600 Hz and 700 Hz, while the upper -3 dB frequency is between 5500 Hz and 6000 Hz. Linear interpolation gives approximately

$$f_{L, \text{exp}} \approx 656 \text{ Hz} \quad f_{H, \text{exp}} \approx 5.755 \text{ kHz}$$

Therefore, the experimental bandwidth and centre frequency are

$$B_{\text{exp}} = f_{H, \text{exp}} - f_{L, \text{exp}} \approx 5.755 - 0.656 \approx 5.099 \text{ kHz}$$

$$f_{c, \text{exp}} = \sqrt{f_{L, \text{exp}} f_{H, \text{exp}}} \approx \sqrt{(656)(5755)} \approx 1.943 \text{ kHz}$$

The experimentally observed peak occurs near 1.9 kHz, which is close to the experimental centre frequency obtained from the cutoff points. The difference between the nominal RC frequencies and the measured cutoff points is consistent with the loading present in a passive cascade.

Observations:

Components used: High-pass section: $R_1 = 1.471 \text{ k}\Omega$, $C_1 = 100.8 \text{ nF}$; Low-pass section: $R_2 = 3.49 \text{ k}\Omega$, $C_2 = 10.65 \text{ nF}$.

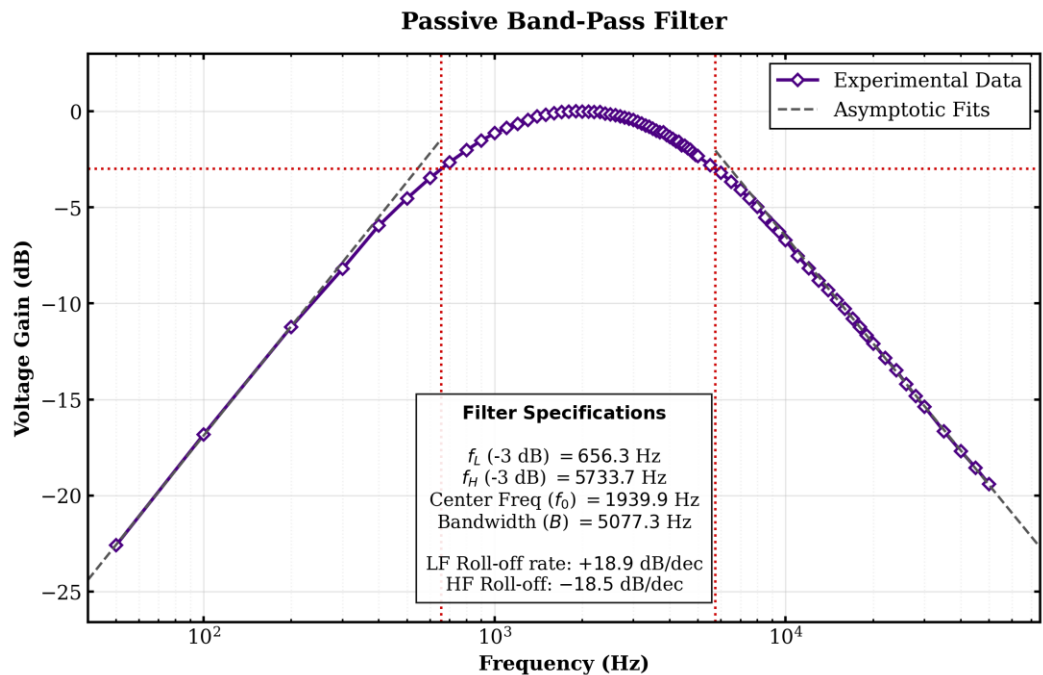
Nominal lower cutoff frequency $\approx 1.073 \text{ kHz}$; Nominal upper cutoff frequency $\approx 4.282 \text{ kHz}$; Input measured voltage, $V_{in} \approx 10 \text{ V}$.

Sr. No.	Frequency (Hz)	Output Voltage (V)	Gain (V_o/V_{in})	Gain(dB)= $20\log_{10}(V_o/V_{in})$
1	50	0.54	0.054	-25.35
2	100	1.05	0.105	-19.58
3	200	2.00	0.200	-13.98
4	300	2.83	0.283	-10.96
5	400	3.67	0.367	-8.71
6	500	4.32	0.432	-7.29
7	600	4.88	0.488	-6.23
8	700	5.36	0.536	-5.42
9	800	5.76	0.576	-4.79
10	900	6.11	0.611	-4.28
11	1,000	6.38	0.638	-3.90
12	1,100	6.59	0.659	-3.62
13	1,200	6.75	0.675	-3.41
14	1,300	6.91	0.691	-3.21
15	1,400	7.06	0.706	-3.02
16	1,500	7.13	0.713	-2.94
17	1,600	7.20	0.720	-2.85
18	1,700	7.24	0.724	-2.81
19	1,800	7.27	0.727	-2.77
20	1,900	7.28	0.728	-2.76
21	2,000	7.27	0.727	-2.77
22	2,100	7.26	0.726	-2.78
23	2,200	7.24	0.724	-2.81
24	2,300	7.24	0.724	-2.81
25	2,400	7.18	0.718	-2.88
26	2,500	7.14	0.714	-2.93
27	2,600	7.11	0.711	-2.96
28	2,700	7.06	0.706	-3.02
29	2,800	7.00	0.700	-3.10
30	2,900	6.96	0.696	-3.15
31	3,000	6.90	0.690	-3.22
32	3,100	6.82	0.682	-3.32
33	3,200	6.76	0.676	-3.40

34	3,300	6.68	0.668	-3.50
35	3,400	6.63	0.663	-3.57
36	3,500	6.56	0.656	-3.66
37	3,600	6.48	0.648	-3.77
38	3,700	6.42	0.642	-3.85
39	3,800	6.41	0.641	-3.86
40	3,900	6.31	0.631	-4.00
41	4,000	6.24	0.624	-4.10
42	4,100	6.16	0.616	-4.21
43	4,200	6.08	0.608	-4.32
44	4,300	6.06	0.606	-4.35
45	4,400	5.94	0.594	-4.52
46	4,500	5.92	0.592	-4.55
47	4,600	5.84	0.584	-4.67
48	4,700	5.78	0.578	-4.76
49	4,800	5.69	0.569	-4.90
50	4,900	5.61	0.561	-5.02
51	5,000	5.56	0.556	-5.10
52	5,500	5.27	0.527	-5.56
53	6,000	5.03	0.503	-5.97
54	6,500	4.77	0.477	-6.43
55	7,000	4.55	0.455	-6.84
56	7,500	4.31	0.431	-7.31
57	8,000	4.10	0.410	-7.74
58	8,500	3.85	0.385	-8.29
59	9,000	3.68	0.368	-8.68
60	9,500	3.53	0.353	-9.04
61	10,000	3.36	0.336	-9.47
62	11,000	3.06	0.306	-10.29
63	12,000	2.84	0.284	-10.93
64	13,000	2.64	0.264	-11.57
65	14,000	2.49	0.249	-12.08
66	15,000	2.35	0.235	-12.58
67	16,000	2.23	0.223	-13.03
68	17,000	2.10	0.210	-13.56
69	18,000	2.00	0.200	-13.98
70	19,000	1.90	0.190	-14.42
71	20,000	1.81	0.181	-14.85
72	22,000	1.66	0.166	-15.60
73	24,000	1.54	0.154	-16.25
74	26,000	1.42	0.142	-16.95
75	28,000	1.32	0.132	-17.59

76	30,000	1.24	0.124	-18.13
77	35,000	1.07	0.107	-19.41
78	40,000	0.95	0.095	-20.45
79	45,000	0.86	0.086	-21.31

Frequency Response:



Result & Discussion:

The frequency response of the passive band-pass filter was obtained experimentally. The filter passes a range of frequencies while attenuating frequencies below and above the passband. The experimental lower and upper cutoff frequencies were found to be **656.3 Hz** and **5733.7 Hz**, respectively. The experimental center frequency was **1939.9 Hz**, which is reasonably close to the expected center frequency of **2 kHz** for the specified 1–4 kHz band-pass filter.

However, the measured cutoff frequencies show considerable deviation from the required **1–4 kHz** range. Consequently, the experimental bandwidth of **5077.3 Hz** is significantly higher than the specified **3 kHz**. The measured roll-off rates are approximately **+18.9 dB/decade** at low frequencies and **-18.5 dB/decade** at high frequencies, which are close to the theoretical **±20 dB/decade** expected for first-order sections.

The deviation in cutoff frequencies and bandwidth may be due to **component tolerances, loading effects between the filter section**. Overall, the graph exhibits the expected band-pass characteristics, although the experimental passband is wider than the specified 1–4 kHz range.

Introduction to Operational Amplifiers & Active Filters

An **operational amplifier (op-amp)** is a high-gain electronic device used to amplify and process electrical signals. It is a differential voltage amplifier, which means that its output depends on the voltage difference between its two input terminals. It is designed to perform mathematical operations such as addition, subtraction, integration, differentiation, and filtering.

Ideally, the output voltage is written as:

$$V_o = A_{OL}(V_+ - V_-)$$

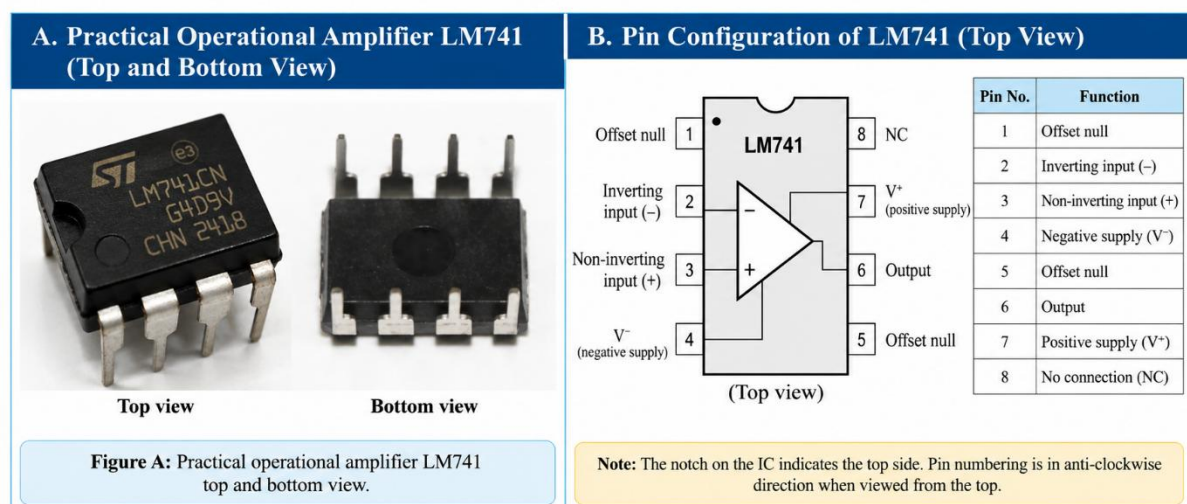
Where,

V_o = output voltage

V_+ = voltage at non-inverting input

V_- = voltage at inverting input

A_{OL} = open-loop voltage gain



Functional Working of Each Pin of IC 741 (LM741):

Pin 1 – Offset Null: Used to adjust the input offset voltage of the op-amp.

Pin 2 – Inverting Input (-): Input applied to this terminal produces an inverted output (180°) when negative feedback is used.

Pin 3 – Non-Inverting Input (+): Input applied to this terminal produces a non-inverted output when negative feedback is used.

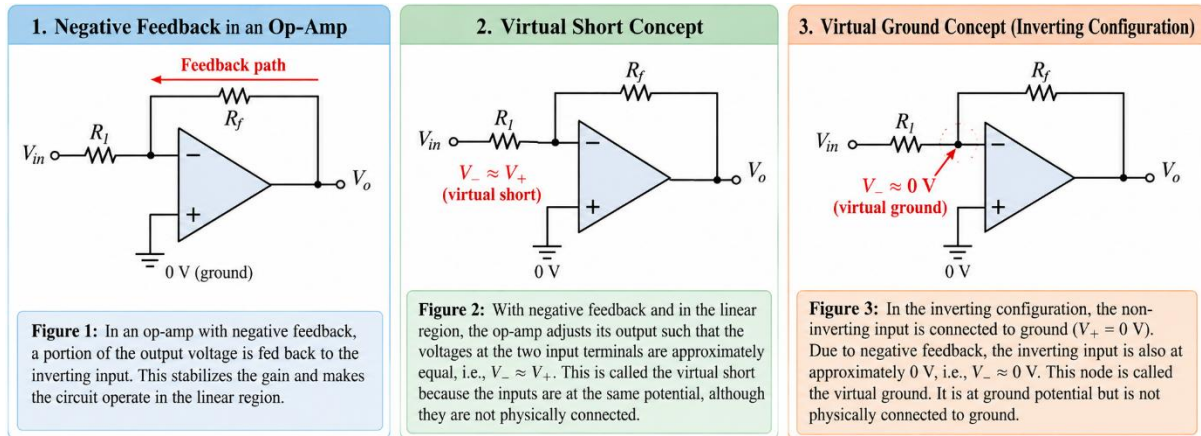
Pin 4 – Negative Supply (V_-): Connected to the negative DC power supply, such as -12 V or -15 V , within the specified operating range.

Pin 5 – Offset Null: Used along with pin 1 to adjust the input offset voltage.

Pin 6 – Output: Provides the amplified or filtered output signal of the op-amp.

Pin 7 – Positive Supply (V+): Connected to the positive DC power supply, such as +12 V or +15 V, within the specified operating range.

Pin 8 – No Connection (NC): Not internally connected to the chip circuitry and is normally left unconnected.



Limitations of Passive Filters:

The passive filters discussed in the previous section are useful for selecting the required frequency components of a signal. However, they have some practical limitations. These limitations become more important when the filter is connected to other circuit stages or when a higher-order frequency response is required.

- 1) **Loading effect:** The cutoff frequency and frequency response of a passive filter can change when a load is connected to its output. The load resistance becomes part of the filter network and changes its effective resistance.
- 2) **No voltage gain:** A passive filter cannot provide voltage amplification. Its output voltage is generally less than the input voltage, so signal attenuation occurs even in the passband. Passive RC filters always exhibit signal loss (gain ≤ 1 , or ≤ 0 dB).
- 3) **Use of inductors:** Higher-order passive filters typically require inductors to obtain sharper frequency selectivity. At audio frequencies (20 Hz – 1 kHz), the required inductances are huge (several Henries). Physical inductors are heavy, bulky, costly, have internal winding resistance, and pick up electromagnetic noise.
- 4) **Poor isolation between stages:** When passive filter stages are connected directly, one stage can affect the operation of another because of loading. This makes it difficult to maintain the designed frequency response.

- 5) **Limited control of gain and frequency response:** In passive circuits, changing component values to modify the frequency response can also affect the signal level and loading conditions.

Need for Active Filters:

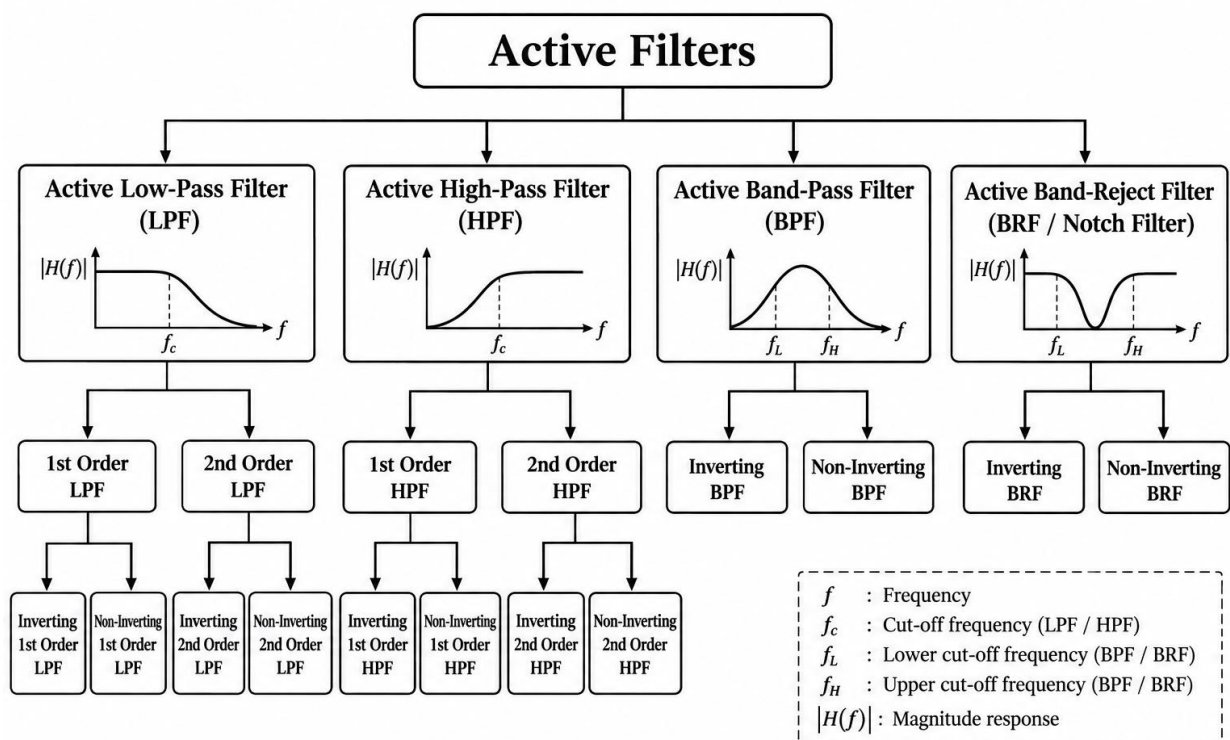
To overcome these limitations, an **operational amplifier (op-amp)** can be combined with resistors and capacitors to form an **active filter**.

An active filter uses:

- a) **Resistors and capacitors** --- determine the frequency-selective behaviour.
- b) **Operational amplifier** --- provides amplification, buffering and isolation.
- c) **Feedback network** --- controls the gain

Main Advantages of Active Filter:

- 1) **High input impedance:** Minimizes loading of the preceding circuit.
- 2) **Low output impedance:** Allows the filter to drive the following circuit effectively.
- 3) **Voltage gain:** Active filters allow the user to filter and simultaneously amplify the signal.
- 4) **Better isolation:** Reduces interaction between cascaded filter stages.
- 5) **No inductors:** Many first- and higher-order filters can be designed using only R , C , and an op-amp.
- 6) **Independent control:** In suitable active-filter configurations, the cutoff frequency and passband gain can be designed with greater independence.



Experiment 1: First Order Active Low Pass Filter (LPF) -

Aim : Design a Non-Inverting First-Order Active Low-Pass Filter and study its frequency response.

Given Parameters:

Gain $A_f = 7$ and Cut off Frequency (f_c) = 1.7 kHz

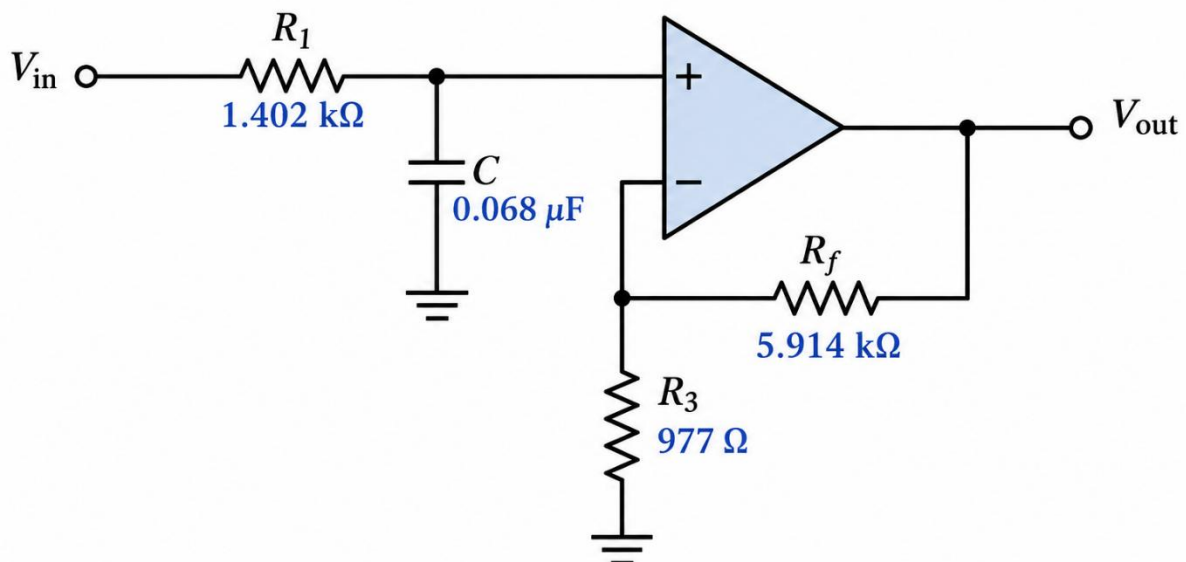
Apparatus Required:

IC 741, Dual track DC power supply ($\pm 12V$ or $\pm 15V$), Function Generator (Sine wave, 10 Hz - 100 kHz), Digital Storage Oscilloscope (DSO) / CRO with probes, Breadboard, Multimeter and single-strand connecting wires

Resistors: $R_1 = (1.183 \text{ k}\Omega + 219.6 \text{ }\Omega \approx 1.402 \text{ k}\Omega)$, $R_f = (5.53 \text{ k}\Omega + 384 \text{ }\Omega \approx 5.914 \text{ k}\Omega)$, $R_3 = 977 \text{ }\Omega$

Capacitor: $C = 0.068 \text{ }\mu\text{F}$

Circuit Diagram:



Non-Inverting Active Low-Pass Filter

Theory:

The input voltage V_{in} is applied to the series resistor R_1 . The capacitor C is connected from the junction of R_1 to the ground. Thus, R_1 and C form a **first-order RC low-pass filter**.

The filtered voltage developed at the junction of R_1 and C is applied to the **non-inverting input** (+) of the op-amp. Since the input impedance of the op-amp is very high, it draws very little current from the RC network, so the loading effect is small.

Gain is given by-

$$A_f = 1 + \frac{R_f}{R_3}$$

At low frequencies, the capacitive reactance is high, so most of the input voltage appears across C . Therefore, the signal is passed to the op-amp and the circuit operates approximately with its passband gain A_f .

As the frequency increases, the capacitive reactance decreases. More of the signal is shunted through the capacitor to ground, so the voltage applied to the non-inverting input decreases. Consequently, the output voltage also decreases.

Therefore, the circuit allows **low-frequency signals to pass while attenuating high-frequency signals**.

The capacitor impedance is-

$$Z_C = \frac{1}{j\omega C}$$

Using voltage division across the capacitor, the voltage at the non-inverting input is-

$$V_+ = V_{in} \times \frac{\frac{1}{j\omega C}}{R_1 + \frac{1}{j\omega C}}$$

Simplifying,

$$V_+ = \frac{V_{in}}{1 + j\omega R_1 C}$$

For the non-inverting amplifier, the voltage at the inverting input is determined by the feedback network R_f and R_3 :

$$V_- = V_{out} \frac{R_3}{R_3 + R_f}$$

Because negative feedback is applied, the virtual-short condition gives-

$$V_+ \approx V_-$$

Therefore,

$$\frac{V_{in}}{1 + j\omega R_1 C} = \frac{R_3}{R_3 + R_f} \times V_{out}$$

Rearranging,

$$\frac{V_{out}}{V_{in}} = \frac{1 + \frac{R_f}{R_3}}{1 + j\omega R_1 C}$$

We know that the gain is - $A_f = 1 + \frac{R_f}{R_3}$ and cut off frequency is- $f_c = \frac{1}{2\pi R_1 C}$

When operating frequency is equal to the cut off frequency, then

$$V_{out} = \frac{A_f \times V_{in}}{\sqrt{2}}$$

So this will come out to be - $\frac{V_{out}}{V_{in}} = \frac{A_f}{1 + j\omega R_1 C}$

Since, $\omega = 2\pi f$, $\omega R_1 C = \frac{f}{f_c}$

So the transfer function changes from-

$$H(j\omega) = \frac{V_{out}}{V_{in}} = \frac{A_f}{1 + j\frac{f}{f_c}}$$

Magnitude of gain –

$$|A_v| = \frac{A_f}{\sqrt{1 + \left(\frac{f}{f_c}\right)^2}}$$

Gain in decibels: $G_{dB} = 20 \log_{10} |A_v|$

Phase shift: $\phi = -\tan^{-1} \left(\frac{f}{f_c} \right)$

$f \ll f_c$: The signal is in the passband, $|A_v| \approx A_f$, and $\phi \approx 0^\circ$.

$f = f_c$: The gain becomes $0.707A_f$, corresponding to -3.0 dB relative to the passband.

$f \gg f_c$: The gain decreases at approximately -20 dB/decade.

Observations:

Resistors: $R_1 = (1.183 \text{ k}\Omega + 219.6 \text{ }\Omega \approx 1.402 \text{ k}\Omega)$, $R_f = (5.53 \text{ k}\Omega + 384 \text{ }\Omega \approx 5.914 \text{ k}\Omega)$, $R_3 = 977 \text{ }\Omega$

Capacitor: $C = 0.068 \text{ }\mu\text{F}$

$$f_c = \frac{1}{2\pi R_1 C} = \frac{1}{2\pi (1.402 \times 10^3)(0.068 \times 10^{-6})} \approx 1.67 \times 10^3 \text{ Hz} = \mathbf{1.67 \text{ kHz}}$$

$$A_f = 1 + \frac{R_f}{R_3} = 1 + \frac{5.914 \times 10^3}{977} \approx \mathbf{7.05}, \quad G_{dB} = 20 \log_{10}(A_f) = 20 \log_{10}(7.05) \approx \mathbf{16.96 \text{ dB}}$$

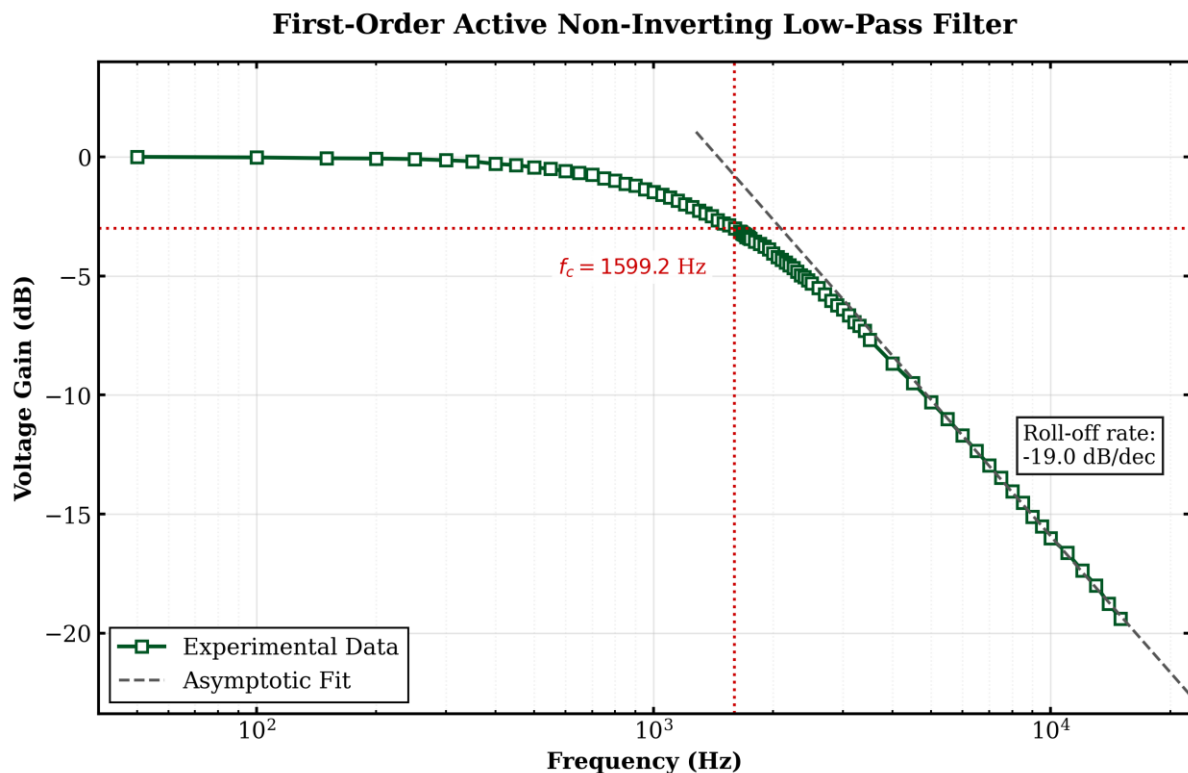
Input voltage : 1 Volt

Sr. No.	Frequency (Hz)	Output Voltage (V)	Gain (V_o/V_{in})	Gain(dB)= $20\log_{10}(V_o/V_{in})$
1	50	7.46	7.46	17.45
2	100	7.44	7.44	17.43
3	150	7.41	7.41	17.4
4	200	7.4	7.40	17.38
5	250	7.38	7.38	17.36
6	300	7.34	7.34	17.31
7	350	7.3	7.30	17.27
8	400	7.21	7.21	17.16
9	450	7.17	7.17	17.11
10	500	7.09	7.09	17.01
11	550	7.04	7.04	16.95
12	600	6.96	6.96	16.85
13	650	6.9	6.90	16.78
14	700	6.84	6.84	16.7
15	750	6.72	6.72	16.55
16	800	6.66	6.66	16.47
17	850	6.55	6.55	16.32
18	900	6.49	6.49	16.24
19	950	6.38	6.38	16.1
20	1,000	6.3	6.30	15.99
21	1,050	6.21	6.21	15.86
22	1,100	6.13	6.13	15.75
23	1,150	6.04	6.04	15.62
24	1,200	5.93	5.93	15.46
25	1,250	5.85	5.85	15.34
26	1,300	5.75	5.75	15.19
27	1,350	5.68	5.68	15.09
28	1,400	5.6	5.60	14.96

29	1,450	5.49	5.49	14.79
30	1,500	5.41	5.41	14.66
31	1,550	5.36	5.36	14.58
32	1,600	5.28	5.28	14.45
33	1,650	5.2	5.20	14.32
34	1,660	5.19	5.19	14.3
35	1,670	5.16	5.16	14.25
36	1,680	5.14	5.14	14.22
37	1,690	5.13	5.13	14.2
38	1,700	5.11	5.11	14.17
39	1,710	5.06	5.06	14.08
40	1,720	5.08	5.08	14.12
41	1,730	5.06	5.06	14.08
42	1,740	5.04	5.04	14.05
43	1,750	5.02	5.02	14.01
44	1,800	4.95	4.95	13.89
45	1,850	4.9	4.90	13.8
46	1,900	4.84	4.84	13.7
47	1,950	4.77	4.77	13.57
48	2,000	4.68	4.68	13.4
49	2,050	4.6	4.60	13.26
50	2,100	4.54	4.54	13.14
51	2,150	4.48	4.48	13.03
52	2,200	4.42	4.42	12.91
53	2,250	4.36	4.36	12.79
54	2,300	4.28	4.28	12.63
55	2,350	4.21	4.21	12.49
56	2,400	4.17	4.17	12.4
57	2,450	4.11	4.11	12.28
58	2,500	4.05	4.05	12.15
59	2,600	3.96	3.96	11.95
60	2,700	3.84	3.84	11.69
61	2,800	3.72	3.72	11.41
62	2,900	3.64	3.64	11.22
63	3,000	3.57	3.57	11.05
64	3,100	3.47	3.47	10.81
65	3,200	3.36	3.36	10.53
66	3,300	3.3	3.30	10.37
67	3,400	3.22	3.22	10.16
68	3,500	3.08	3.08	9.77
69	4,000	2.75	2.75	8.79
70	4500	2.5	2.5	7.96
71	5,000	2.28	2.28	7.16

72	5,500	2.1	2.10	6.44
73	6,000	1.94	1.94	5.76
74	6,500	1.8	1.80	5.11
75	7,000	1.68	1.68	4.51
76	7,500	1.58	1.58	3.97
77	8,000	1.48	1.48	3.41
78	8500	1.4	1.4	2.92
79	9,000	1.31	1.31	2.35
80	9,500	1.25	1.25	1.94
81	10,000	1.18	1.18	1.44
82	11,000	1.1	1.10	0.83
83	12,000	1.01	1.01	0.09
84	13000	0.94	0.94	-0.54
85	14,000	0.86	0.86	-1.31
86	15,000	0.8	0.80	-1.94

Frequency Response:



Result & Discussion:

The frequency response of the first-order active non-inverting low-pass filter was obtained experimentally. The filter shows nearly constant gain at low frequencies and attenuation as the frequency increases. The experimental cutoff frequency was found to be approximately **1599.2**

Hz, at which the gain is about **-3 dB**. The measured roll-off rate is approximately **-19 dB/decade**, which is close to the theoretical value of **-20 dB/decade** for a first-order low-pass filter. The small deviation may be due to component tolerances, op-amp limitations, and measurement errors. Thus, the experimental response agrees well with the expected low-pass filter characteristics.

Experiment 2: First Order Active Low Pass Filter (LPF) -

Aim : Design an Inverting First-Order Active Low-Pass Filter and study its frequency response.

Given Parameters:

Gain $A_f = 5$ and Cut off Frequency (f_c) = 1.7 kHz

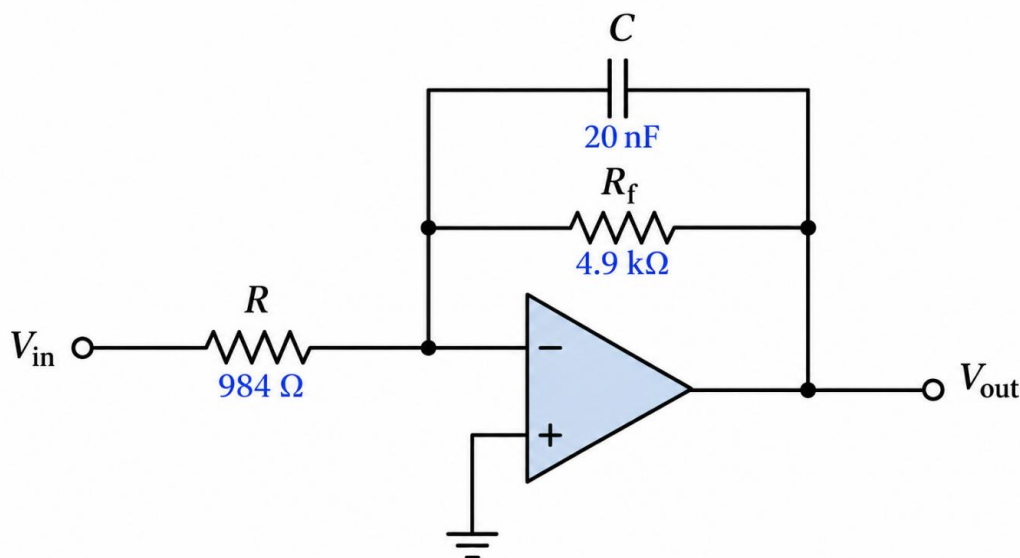
Apparatus Required:

IC 741, Dual track DC power supply ($\pm 12V$ or $\pm 15V$), Function Generator (Sine wave, 10 Hz - 100 kHz), Digital Storage Oscilloscope (DSO) / CRO with probes, Breadboard, Multimeter and single-strand connecting wires

Resistors: $R = 984 \Omega$, $R_f = (4.68 \text{ k}\Omega + 219 \Omega)$

Capacitor: $C = 0.020 \mu\text{F}$ or 20 nF

Circuit Diagram:



Inverting Active Low-Pass Filter

Theory:

In the inverting active low-pass filter, the input signal V_{in} is applied to the **inverting input** (–) of the op-amp through an input resistor R . A feedback network consisting of a resistor R_f and capacitor C is connected between the output and the inverting input.

The non-inverting input (+) is connected to ground. Due to negative feedback, the inverting input is maintained at approximately 0 V , giving the **virtual-ground condition**.

The resistor R together with the feedback network determines the frequency-dependent gain of the circuit.

At **low frequencies**, the capacitor has high impedance and behaves approximately as an open circuit. Therefore, the feedback is mainly through R_f , and the circuit has approximately constant inverting gain:

$$A_f = -\frac{R_f}{R}$$

As the frequency increases, the impedance of the capacitor decreases. The capacitor therefore provides an increasingly effective path in the feedback network, reducing the magnitude of the closed-loop gain.

Thus, the circuit **passes low-frequency signals and attenuates high-frequency signals**. Since the signal is applied to the inverting input, the output is **180° out of phase** with the input in the passband.

Observations:

Resistors: $R = 984\ \Omega$, $R_f = (4.68\text{ k}\Omega + 219\ \Omega)$, Capacitor: $C = 0.020\ \mu\text{F}$ or 20 nF

$$f_c = \frac{1}{2\pi(4.9 \times 10^3)(20 \times 10^{-9})} \approx \mathbf{1.62\text{ kHz}}, \quad A_f = -\frac{4.9 \times 10^3}{984} \approx \mathbf{-4.98}$$

$$G_{dB} = 20 \log_{10}(4.98) \approx \mathbf{13.94\text{ dB}}$$

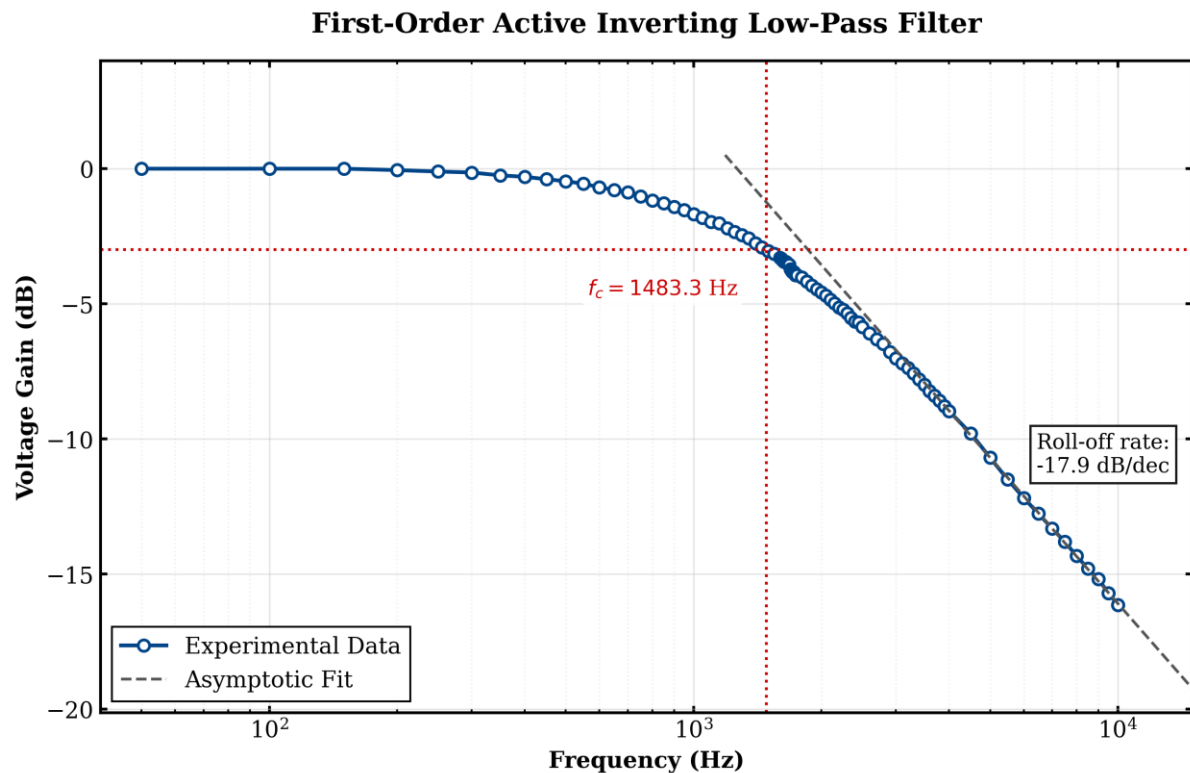
Input voltage : 1 Volt

Sr. No.	Frequency (Hz)	Output Voltage (V)	Gain (V_o/V_{in})	Gain(dB)= $20\log_{10}(V_o/V_{in})$
1	50	5	5.00	13.98
2	100	5	5.00	13.98
3	150	5	5.00	13.98
4	200	4.97	4.97	13.93
5	250	4.94	4.94	13.87

6	300	4.92	4.92	13.84
7	350	4.86	4.86	13.73
8	400	4.83	4.83	13.68
9	450	4.78	4.78	13.59
10	500	4.74	4.74	13.52
11	550	4.69	4.69	13.42
12	600	4.62	4.62	13.29
13	650	4.57	4.57	13.2
14	700	4.52	4.52	13.1
15	750	4.44	4.44	12.95
16	800	4.37	4.37	12.81
17	850	4.32	4.32	12.71
18	900	4.25	4.25	12.57
19	950	4.19	4.19	12.44
20	1,000	4.12	4.12	12.3
21	1,050	4.06	4.06	12.17
22	1,100	3.99	3.99	12.02
23	1,150	3.96	3.96	11.95
24	1,200	3.88	3.88	11.78
25	1,250	3.82	3.82	11.64
26	1,300	3.77	3.77	11.53
27	1,350	3.72	3.72	11.41
28	1,400	3.64	3.64	11.22
29	1,450	3.58	3.58	11.08
30	1,500	3.52	3.52	10.93
31	1,550	3.48	3.48	10.83
32	1,600	3.42	3.42	10.68
33	1,610	3.41	3.41	10.66
34	1,620	3.39	3.39	10.6
35	1,630	3.37	3.37	10.55
36	1,640	3.36	3.36	10.53
37	1,650	3.36	3.36	10.53
38	1,660	3.35	3.35	10.5
39	1,670	3.33	3.33	10.45
40	1,680	3.32	3.32	10.42
41	1,690	3.26	3.26	10.26
42	1,700	3.24	3.24	10.21
43	1,710	3.22	3.22	10.16
44	1,720	3.22	3.22	10.16
45	1,730	3.2	3.20	10.1
46	1,740	3.18	3.18	10.05
47	1,750	3.18	3.18	10.05
48	1,800	3.15	3.15	9.97

49	1,850	3.09	3.09	9.8
50	1,900	3.04	3.04	9.66
51	1,950	2.99	2.99	9.51
52	2,000	2.95	2.95	9.4
53	2,050	2.91	2.91	9.28
54	2,100	2.86	2.86	9.13
55	2,150	2.82	2.82	9
56	2,200	2.77	2.77	8.85
57	2,250	2.74	2.74	8.76
58	2,300	2.7	2.70	8.63
59	2,350	2.65	2.65	8.46
60	2,400	2.61	2.61	8.33
61	2,450	2.6	2.60	8.3
62	2,500	2.55	2.55	8.13
63	2,600	2.48	2.48	7.89
64	2,700	2.42	2.42	7.68
65	2,800	2.37	2.37	7.49
66	2,900	2.29	2.29	7.2
67	3,000	2.23	2.23	6.97
68	3,100	2.18	2.18	6.77
69	3,200	2.14	2.14	6.61
70	3,300	2.09	2.09	6.4
71	3,400	2.04	2.04	6.19
72	3,500	1.99	1.99	5.98
73	3,600	1.94	1.94	5.76
74	3,700	1.9	1.90	5.58
75	3,800	1.86	1.86	5.39
76	3900	1.82	1.82	5.2
77	4,000	1.78	1.78	5.01
78	4,500	1.62	1.62	4.19
79	5,000	1.46	1.46	3.29
80	5,500	1.33	1.33	2.48
81	6,000	1.23	1.23	1.8
82	6,500	1.15	1.15	1.21
83	7,000	1.08	1.08	0.67
84	7500	1.02	1.02	0.17
85	8,000	0.96	0.96	-0.35
86	8,500	0.91	0.91	-0.82
87	9,000	0.87	0.87	-1.21
88	9,500	0.82	0.82	-1.72
89	10,000	0.78	0.78	-2.16

Frequency Response:



Result and Discussion:

The frequency response of the first-order active inverting low-pass filter was obtained experimentally. The filter shows nearly constant gain at low frequencies and attenuation as the frequency increases. The experimental cutoff frequency was found to be approximately **1483.3 Hz**, at which the gain is about **-3 dB**. The measured roll-off rate is approximately **-17.9 dB/decade**, which is reasonably close to the theoretical value of **-20 dB/decade** for a first-order low-pass filter. The small deviation may be due to component tolerances, op-amp limitations, and measurement errors. Thus, the experimental response agrees reasonably well with the expected characteristics of a first-order active inverting low-pass filter.

Experiment 3: Second Order Active Low Pass Filter (LPF):

Aim : Design a Non Inverting Second-Order Active Low-Pass Filter and study its frequency response.

Given Parameters:

Gain $A_f = 7$ and Cut off Frequency (f_c) = 1.7 kHz

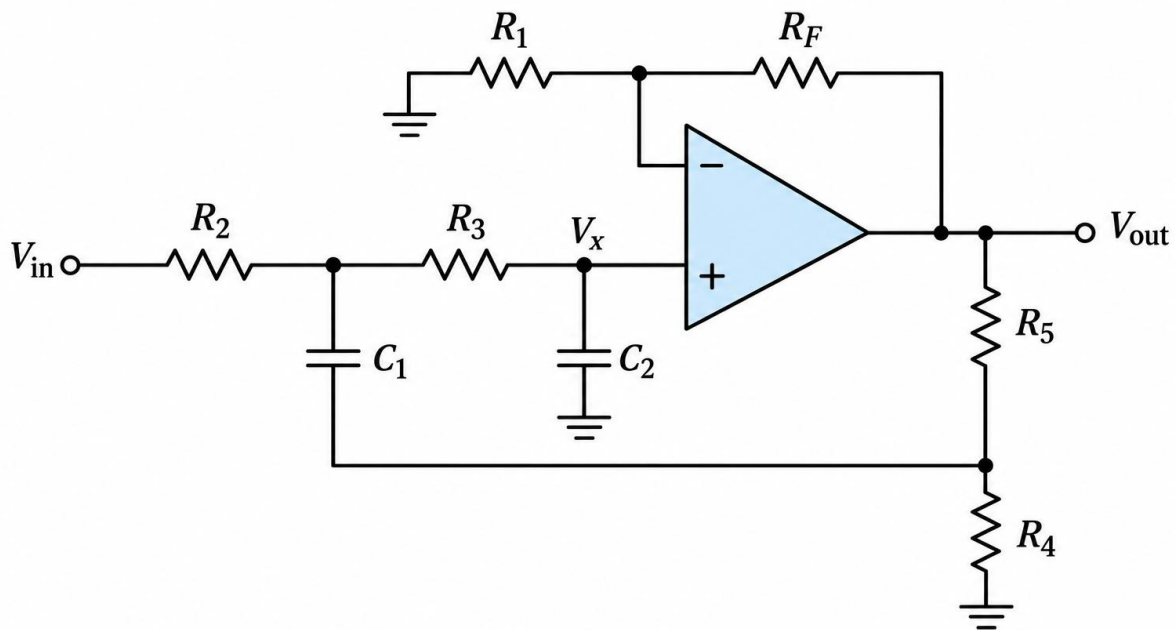
Apparatus Required:

IC 741, Dual track DC power supply ($\pm 12V$ or $\pm 15V$), Function Generator (Sine wave, 10 Hz - 100 kHz), Digital Storage Oscilloscope (DSO) / CRO with probes, Breadboard, Multimeter and single-strand connecting wires

*****Resistors: $R_1 = (1.473 \text{ k}\Omega + 546 \text{ }\Omega \approx 2.019 \text{ k}\Omega)$, $R_2 = 977 \text{ }\Omega$, $R_f = (5.53 \text{ k}\Omega + 384 \text{ }\Omega \approx 5.914 \text{ k}\Omega)$

*****Capacitor: $C_1 = 0.025 \text{ }\mu\text{F}$ or 25 nF

Circuit Diagram:



2nd Order Non-Inverting Low-Pass Filter

Theory:

A second order Active Low-Pass Filter (LPF) passes signal frequencies lower than a selected cutoff frequency (f_c) while attenuating signals above f_c . It has an asymptotic attenuation slope of -40 dB/decade, which is equivalent to approximately -12 dB/octave.

Butterworth Filter Characteristics

A second-order Butterworth filter provides a maximally flat magnitude response in the passband with no ripple. Its standard second-order parameters are:

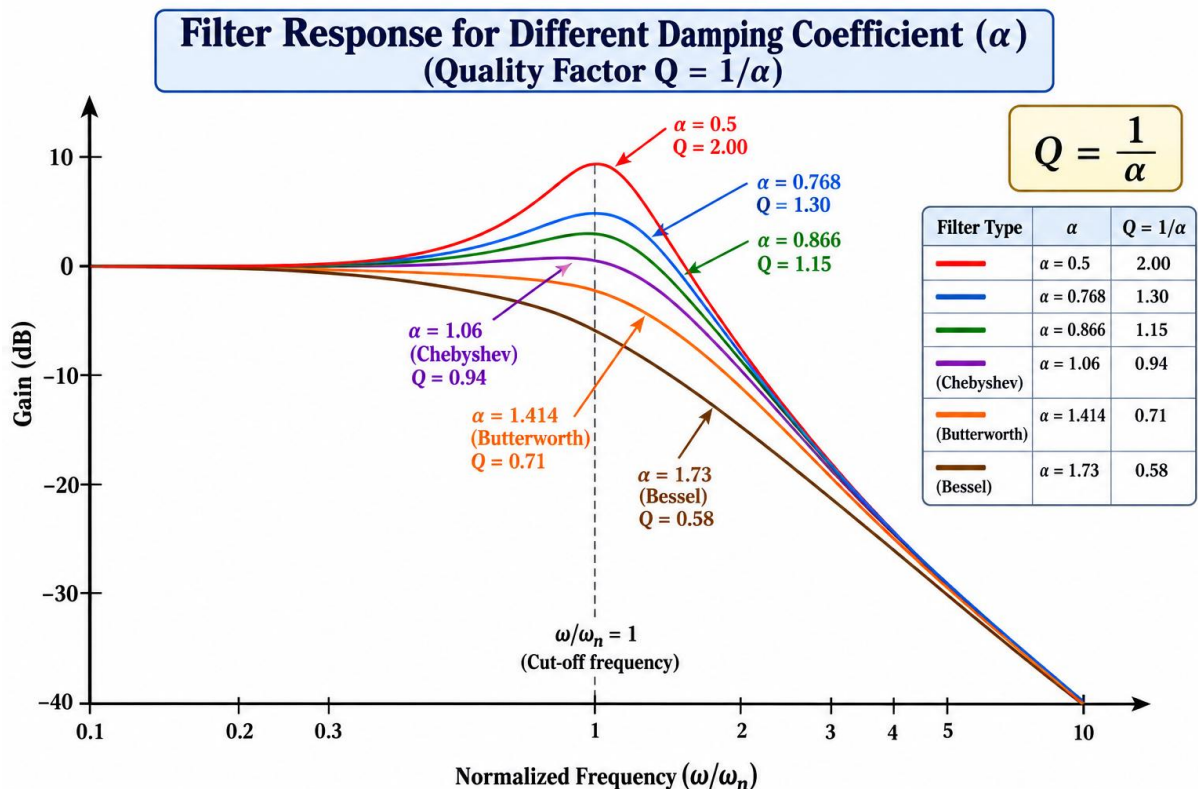
$$\zeta = \frac{1}{2Q} \approx 0.707, \quad Q = \frac{1}{\sqrt{2}}$$

Here ζ is Damping ratio and Q is Quality factor.

Standard Sallen-Key Topology & the Gain Stability Limitation

In a conventional non-inverting Sallen-Key low-pass filter:

- ❖ R_2 and R_3 are the series resistors in the signal path.
- ❖ C_2 connects the non-inverting terminal (+) to ground.
- ❖ C_1 connects the intermediate signal node directly to the op-amp output (V_{out}).



The passband gain K is established by the op-amp's negative-feedback resistors.

$$K = 1 + \frac{R_f}{R_1}$$

For equal components, $R_2 = R_3 = R$ and $C_1 = C_2 = C$, the conventional Sallen-Key transfer function becomes:

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{K}{(sRC)^2 + (3 - K)sRC + 1}$$

$$H(s) = \frac{K}{s^2 R^2 C^2 + (3 - K)sRC + 1}$$

Divide by $R^2 C^2$:

$$H(s) = \frac{\frac{K}{R^2 C^2}}{s^2 + \frac{3 - K}{RC}s + \frac{1}{R^2 C^2}}$$

The standard form of 2nd order LPF is :

$$H(s) = \frac{K\omega_n^2}{s^2 + \left(\frac{\omega_n}{Q}\right)s + \omega_n^2}$$

Now on comparing denominator coefficient of s^2 and constant term of both $H(s)$:

$$f_c = \frac{\omega_n}{2\pi} = \frac{1}{2\pi RC}$$

$$Q = \frac{1}{3 - K}$$

$$K = 3 - \sqrt{2} \approx 1.586$$

The high-gain problem ($K \geq 3$)

For a Butterworth response, Q must be approximately 0.707. With equal R and C values, the required gain is therefore:

$$K = 3 - \sqrt{2} \approx 1.586, \quad Q = \frac{1}{3 - K}$$

If K approaches 3, the damping term $(3 - K)$ approaches zero and Q becomes very large, producing strong resonant peaking. If K exceeds 3, the damping term becomes negative and

the idealized denominator has an unstable right-half-plane pole pair. Therefore, a standard equal-component Sallen-Key filter cannot be used directly to obtain a large passband gain such as $K = 7$ while maintaining a Butterworth response.

$$1 + \frac{R_f}{R_1} = 1.586$$

$$\frac{R_f}{R_1} = 1.586 - 1 = 0.586$$

$$R_f = 0.586R_1$$

Modified Sallen-Key Topology (Feedback Attenuation)

To obtain a high passband gain (for example, $K = 7$), we can select R_1 , R_2 , C_1 , and C_2 as different values. However, this makes the design and calculations more complex. To reduce this complexity, the circuit is modified by introducing a voltage divider for positive-feedback attenuation, while retaining equal passive components:

$$R_2 = R_3 = R, \quad C_1 = C_2 = C$$

R_5 is connected between the op-amp output (V_{out}) and the feedback junction.

R_4 is connected between the feedback junction and ground.

C_1 is connected to this intermediate feedback junction rather than directly to V_{out} .

Hence the resistor divider produces an attenuation factor x :

Therefore, the voltage at the feedback junction is:

$$x = \frac{R_4}{R_4 + R_5}, \quad \frac{R_4}{R_4 + R_5} = \frac{3 - \sqrt{2}}{K}$$

The capacitor consequently receives only a fraction x of the output voltage. The effective positive-feedback contribution is therefore governed by xK rather than K alone. The damping term becomes $(3 - xK)$, allowing the overall op-amp gain K to be large provided x is selected appropriately.

For example, if $K = 7$:

$$x = \frac{1.586}{7} \approx 0.2266; \quad x < 1$$

The divider must attenuate the output to approximately 22.66% at the C₁ feedback junction. Equivalently:

$$\frac{R_5}{R_4} = \frac{1 - 0.2266}{0.2266} \approx 3.413$$

$$R_5 = 3.413 \times R_4$$

Now feedback resistance is:

$$1 + \frac{R_f}{R_1} = 7, \quad \frac{R_f}{R_1} = 7 - 1 = 6$$

$$R_f = 6R_1$$

#####

Observations:

Resistors: R₁ = (1.473 kΩ + 546 Ω ≈ 2.019 kΩ), R₂ = 977 Ω, R_f = (5.53 kΩ + 384 Ω ≈ 5.914 kΩ), Capacitor: C₁ = 0.025 μF or 25 nF

$$f_c = \frac{1}{2\pi(2.019 \times 10^3)(25 \times 10^{-9})} \approx 3.15 \text{ kHz}, \quad A_f = 1 + \frac{5.914 \times 10^3}{977} \approx 7.05$$

$$G_{dB} = 20 \log_{10}(7.05) \approx 16.96 \text{ dB}$$

Frequency Response:

#####

Experiment 4: First Order Active High Pass Filter (HPF) -

Aim : Design a Non Inverting First-Order Active High-Pass Filter and study its frequency response.

Given Parameters:

Gain $A_f = 7$ and Cut off Frequency (f_c) = 3.2 kHz

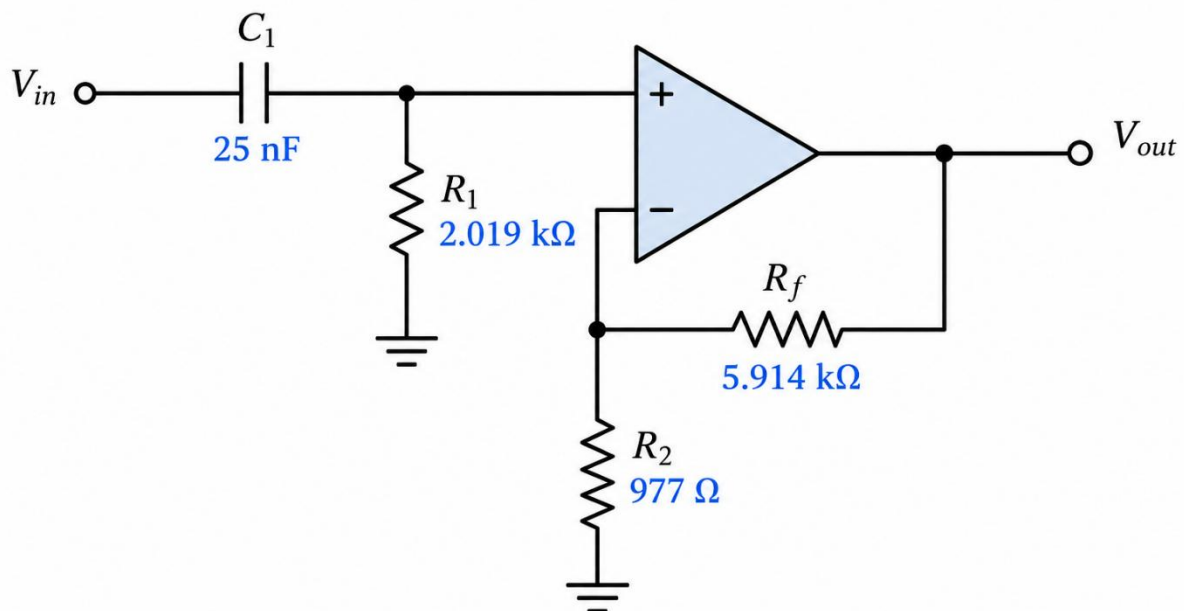
Apparatus Required:

IC 741, Dual track DC power supply ($\pm 12V$ or $\pm 15V$), Function Generator (Sine wave, 10 Hz - 100 kHz), Digital Storage Oscilloscope (DSO) / CRO with probes, Breadboard, Multimeter and single-strand connecting wires

Resistors: $R_1 = (1.473 \text{ k}\Omega + 546 \text{ }\Omega \approx 2.019 \text{ k}\Omega)$, $R_2 = 977 \text{ }\Omega$, $R_f = (5.53 \text{ k}\Omega + 384 \text{ }\Omega \approx 5.914 \text{ k}\Omega)$

Capacitor: $C_1 = 0.025 \text{ }\mu\text{F}$ or 25 nF

Circuit Diagram:



Non-Inverting Active High-Pass Filter

Theory:

A first-order active high-pass filter allows signals with frequencies **above the cutoff frequency** to pass while attenuating lower-frequency components. The circuit combines a passive RC high-pass network with a non-inverting operational amplifier. In this circuit, the input signal is first applied to the **series combination of C_1 and R_1** . The capacitor C_1 is responsible for the high-pass action. At very low frequencies, the reactance of the capacitor is high, so only a small portion of the input signal appears at the non-inverting input of the op-amp. As the frequency increases, the capacitive reactance decreases, allowing a larger portion of the input signal to reach the non-inverting input.

The combination of R_1 and C_1 therefore determines the **cutoff frequency** of the filter. Below the cutoff frequency, the output amplitude is strongly attenuated, while above the cutoff frequency, the input signal is passed to the op-amp with much less attenuation.

$$f_c = \frac{1}{2\pi R_1 C_1}, \quad A_f = 1 + \frac{R_f}{R_2}$$

Transfer function is given by-

$$H(j\omega) = \frac{V_{out}}{V_{in}} = \frac{A_f(j\omega R_1 C_1)}{1 + j\omega R_1 C_1}$$

The op-amp is connected in the **non-inverting configuration**. Negative feedback through R_f and R_2 sets the voltage gain of the amplifier. Under negative feedback, the voltages at the two input terminals are approximately equal, so the voltage developed at the non-inverting input is effectively amplified by the closed-loop gain of the op-amp.

Observations:

Resistors: $R_1 = (1.473 \text{ k}\Omega + 546 \text{ }\Omega \approx 2.019 \text{ k}\Omega)$, $R_2 = 977 \text{ }\Omega$, $R_f = (5.53 \text{ k}\Omega + 384 \text{ }\Omega \approx 5.914 \text{ k}\Omega)$, Capacitor: $C_1 = 0.025 \text{ }\mu\text{F}$ or 25 nF

$$f_c = \frac{1}{2\pi(2.019 \times 10^3)(25 \times 10^{-9})} \approx \mathbf{3.15 \text{ kHz}}, \quad A_f = 1 + \frac{5.914 \times 10^3}{977} \approx \mathbf{7.05}$$

$$G_{dB} = 20 \log_{10}(7.05) \approx \mathbf{16.96 \text{ dB}}$$

Input voltage : 1 Volt

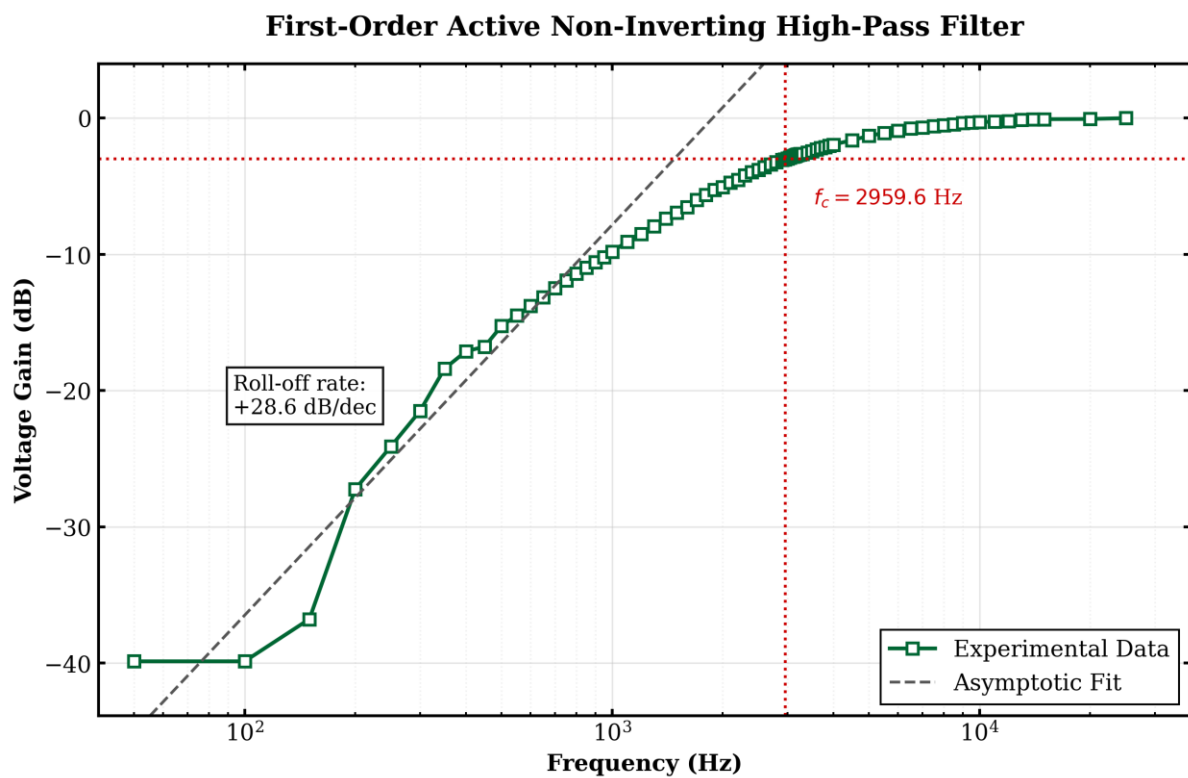
Sr. No.	Frequency (Hz)	Output Voltage (V)	Gain (V_o/V_{in})	Gain(dB)= $20\log_{10}(V_o/V_{in})$
1	50	0.07	0.07	-23.1
2	100	0.07	0.07	-23.1
3	150	0.10	0.10	-20

4	200	0.30	0.30	-10.46
5	250	0.43	0.43	-7.33
6	300	0.58	0.58	-4.73
7	350	0.83	0.83	-1.62
8	400	0.96	0.96	-0.35
9	450	1.00	1.00	0
10	500	1.19	1.19	1.51
11	550	1.30	1.30	2.28
12	600	1.41	1.41	2.98
13	650	1.52	1.52	3.64
14	700	1.64	1.64	4.3
15	750	1.75	1.75	4.86
16	800	1.85	1.85	5.34
17	850	1.95	1.95	5.8
18	900	2.04	2.04	6.19
19	950	2.13	2.13	6.57
20	1,000	2.23	2.23	6.97
21	1,100	2.43	2.43	7.71
22	1,200	2.59	2.59	8.27
23	1,300	2.76	2.76	8.82
24	1,400	2.95	2.95	9.4
25	1,500	3.10	3.10	9.83
26	1,600	3.25	3.25	10.24
27	1,700	3.46	3.46	10.78
28	1,800	3.60	3.60	11.13
29	1,900	3.76	3.76	11.5
30	2,000	3.85	3.85	11.71
31	2,100	4.00	4.00	12.04
32	2,200	4.09	4.09	12.23
33	2,300	4.26	4.26	12.59
34	2,400	4.36	4.36	12.79
35	2,500	4.46	4.46	12.99
36	2,600	4.55	4.55	13.16
37	2,700	4.67	4.67	13.39
38	2,800	4.75	4.75	13.53
39	2,900	4.84	4.84	13.7
40	2,950	4.87	4.87	13.75
41	3,000	4.91	4.91	13.82
42	3,050	4.94	4.94	13.87
43	3,100	4.98	4.98	13.94

44	3,110	4.99	4.99	13.96
45	3,120	4.99	4.99	13.96
46	3,130	5.00	5.00	13.98
47	3,140	5.01	5.01	14
48	3,150	5.01	5.01	14
49	3,160	5.03	5.03	14.03
50	3,170	5.04	5.04	14.05
51	3,180	5.05	5.05	14.07
52	3,190	5.05	5.05	14.07
53	3,200	5.06	5.06	14.08
54	3,210	5.07	5.07	14.1
55	3,220	5.08	5.08	14.12
56	3,230	5.08	5.08	14.12
57	3,240	5.08	5.08	14.12
58	3,250	5.09	5.09	14.13
59	3,300	5.10	5.10	14.15
60	3,400	5.16	5.16	14.25
61	3,500	5.21	5.21	14.34
62	3,600	5.29	5.29	14.47
63	3,700	5.35	5.35	14.57
64	3,800	5.40	5.40	14.65
65	3,900	5.45	5.45	14.73
66	4,000	5.51	5.51	14.82
67	4,500	5.72	5.72	15.15
68	5,000	5.93	5.93	15.46
69	5,500	6.08	6.08	15.68
70	6,000	6.19	6.19	15.83
71	6,500	6.31	6.31	16
72	7,000	6.37	6.37	16.08
73	7,500	6.45	6.45	16.19
74	8,000	6.50	6.50	16.26
75	8,500	6.54	6.54	16.31
76	9,000	6.61	6.61	16.4
77	9,500	6.63	6.63	16.43
78	10,000	6.66	6.66	16.47
79	11,000	6.69	6.69	16.51
80	12,000	6.73	6.73	16.56
81	13,000	6.80	6.80	16.65
82	14,000	6.81	6.81	16.66
83	15,000	6.83	6.83	16.69

84	20,000	6.84	6.84	16.7
85	25,000	6.89	6.89	16.76
86	30,000	6.63	6.63	16
87	35,000	6.30	6.30	15.99
88	40,000	5.82	5.82	15.3
89	45,000	5.36	5.36	14.58
90	50,000	4.94	4.94	13.87
91	60,000	4.21	4.21	12.49
92	70,000	3.66	3.66	11.27
93	80,000	3.17	3.17	10.02
94	90,000	2.81	2.81	8.97
95	1,00,000	2.56	2.56	8.16
96	1,50,000	1.73	1.73	4.76
97	2,00,000	1.28	1.28	2.14
98	2,50,000	1.03	1.03	0.26
99	3,00,000	0.88	0.88	-1.11

Frequency Response:



Result and Discussion:

The frequency response of the first-order active non-inverting high-pass filter was obtained experimentally. At low frequencies, the gain is strongly attenuated, while the gain increases with frequency. The experimental cutoff frequency is approximately **2599.6 Hz**, at which the gain is about **-3 dB**. At higher frequencies, the gain approaches a nearly constant passband value of **0 dB**.

An **unusual behavior** is observed in the low-frequency region, where the gain remains almost constant at around **-40 dB** before beginning to rise. Also, the measured roll-off rate is approximately **+28.6 dB/decade**, which is higher than the theoretical **+20 dB/decade** expected for a first-order high-pass filter. This deviation can be attributed to the limited measurement range, finite resolution of the experimental data, component tolerances, and practical limitations of the op-amp and measurement setup.

Overall, the graph shows the expected **high-pass characteristic**, with strong attenuation at low frequencies and nearly constant gain at high frequencies.

Experiment 5: First Order Active High Pass Filter (HPF) -

Aim : Design an Inverting First-Order Active High-Pass Filter and study its frequency response.

Given Parameters:

Gain $A_f = 5$ and Cut off Frequency (f_c) = 3.2 kHz

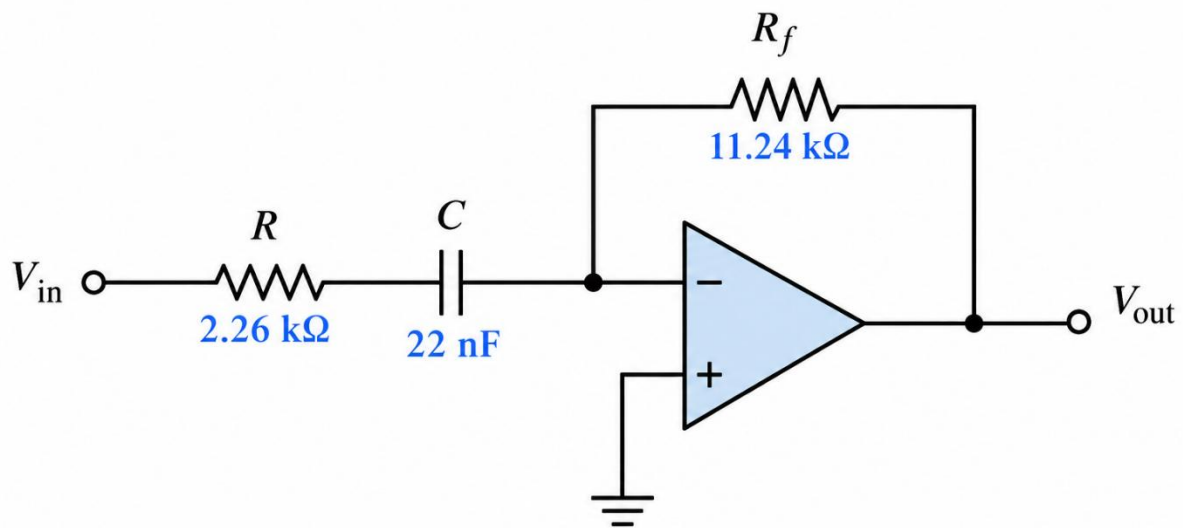
Apparatus Required:

IC 741, Dual track DC power supply ($\pm 12V$ or $\pm 15V$), Function Generator (Sine wave, 10 Hz - 100 kHz), Digital Storage Oscilloscope (DSO) / CRO with probes, Breadboard, Multimeter and single-strand connecting wires

Resistors: $R = 2.26 \text{ k}\Omega$, $R_f = (1.47 \text{ k}\Omega + 9.77 \text{ k}\Omega \approx 11.24 \text{ k}\Omega)$

Capacitor: $C = 0.022 \text{ }\mu\text{F}$ or 22 nF

Circuit Diagram:



Inverting Active High-Pass Filter

Theory:

A first-order active high-pass filter in the **inverting configuration** combines an RC high-pass network with an operational amplifier. The capacitor C is connected in series with the input resistor R , so it blocks DC and attenuates low-frequency signals. As the frequency increases, the capacitive reactance decreases, allowing more of the input signal to reach the inverting input of the op-amp.

The R - C network determines the **cutoff frequency**, while R_f controls the passband gain. Since the non-inverting terminal is grounded and negative feedback is applied, the inverting input is at approximately **virtual ground**.

At frequencies below the cutoff frequency, the output is strongly attenuated. Above the cutoff frequency, the circuit approaches a constant **inverting gain**, producing an output that is 180° out of phase with the input.

$$A_f = -\frac{R_f}{R}, \quad f_c = \frac{1}{2\pi RC}$$

Observations:

Resistors: $R = 2.26 \text{ k}\Omega$, $R_f = (1.47 \text{ k}\Omega + 9.77 \text{ k}\Omega \approx 11.24 \text{ k}\Omega)$, Capacitor: $C = 22 \text{ nF}$

$$f_c = \frac{1}{2\pi(2.26 \times 10^3)(22 \times 10^{-9})} \approx 3.20 \text{ kHz}, \quad A_f = -\frac{11.24}{2.26} \approx -4.97$$

$$G_{dB} = 20 \log_{10}(4.97) \approx 13.93 \text{ dB}$$

Input voltage : 1 Volt

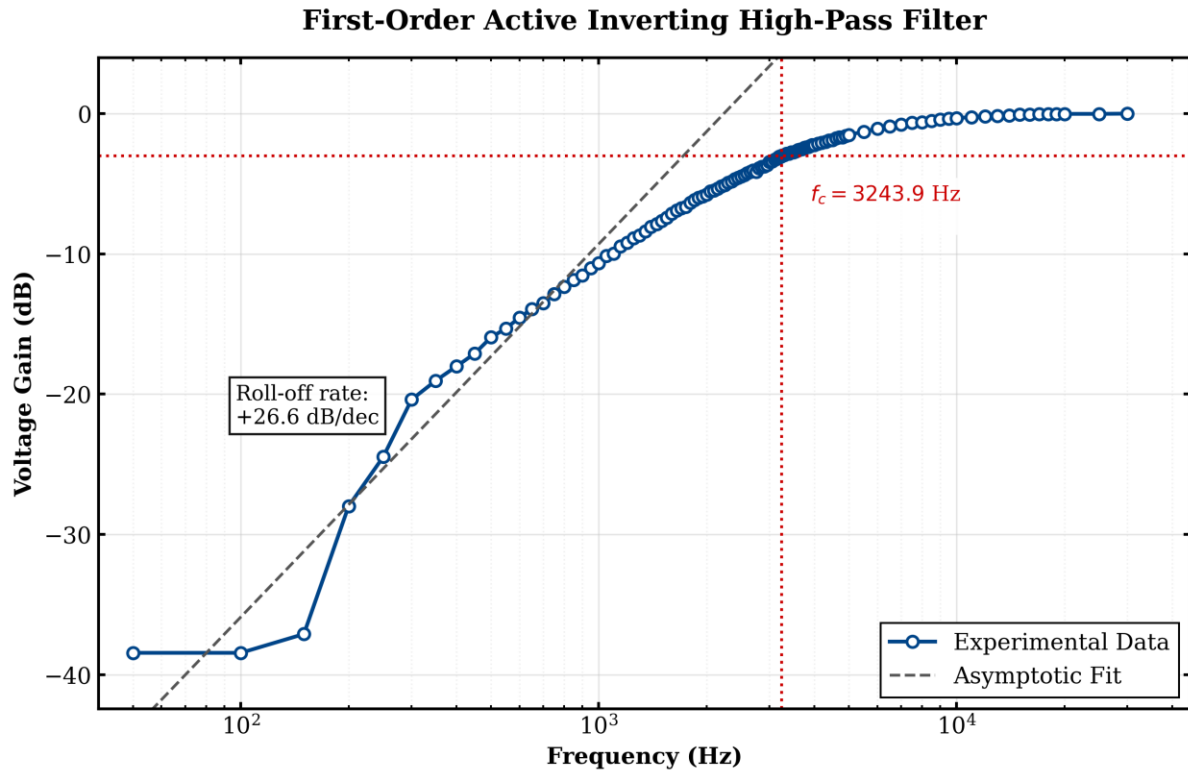
Sr. No.	Frequency (Hz)	Output Voltage (V)	Gain (V_o/V_{in})	Gain(dB)= $20\log_{10}(V_o/V_{in})$
1	50	0.06	0.06	-24.44
2	100	0.06	0.06	-24.44
3	150	0.07	0.07	-23.10
4	200	0.2	0.20	-13.98
5	250	0.3	0.30	-10.46
6	300	0.48	0.48	-6.38
7	350	0.56	0.56	-5.04
8	400	0.63	0.63	-4.01
9	450	0.7	0.70	-3.10
10	500	0.8	0.80	-1.94
11	550	0.86	0.86	-1.31
12	600	0.94	0.94	-0.54
13	650	1.01	1.01	0.09
14	700	1.06	1.06	0.51
15	750	1.14	1.14	1.14
16	800	1.21	1.21	1.66
17	850	1.28	1.28	2.14
18	900	1.33	1.33	2.48
19	950	1.41	1.41	2.98
20	1,000	1.47	1.47	3.35
21	1,050	1.56	1.56	3.86
22	1,100	1.59	1.59	4.03
23	1,150	1.69	1.69	4.56
24	1,200	1.74	1.74	4.81
25	1,250	1.81	1.81	5.15
26	1,300	1.85	1.85	5.34
27	1,350	1.91	1.91	5.62
28	1,400	1.98	1.98	5.93
29	1,450	2.03	2.03	6.15
30	1,500	2.08	2.08	6.36
31	1,550	2.14	2.14	6.61

32	1,600	2.21	2.21	6.89
33	1,650	2.27	2.27	7.12
34	1,700	2.31	2.31	7.27
35	1,750	2.34	2.34	7.38
36	1,800	2.42	2.42	7.68
37	1,850	2.47	2.47	7.85
38	1,900	2.52	2.52	8.03
39	1,950	2.55	2.55	8.13
40	2,000	2.58	2.58	8.23
41	2,050	2.65	2.65	8.46
42	2,100	2.67	2.67	8.53
43	2,150	2.71	2.71	8.66
44	2,200	2.76	2.76	8.82
45	2,250	2.8	2.80	8.94
46	2,300	2.85	2.85	9.10
47	2,350	2.88	2.88	9.19
48	2,400	2.93	2.93	9.34
49	2,450	2.97	2.97	9.46
50	2,500	3	3.00	9.54
51	2,550	3.03	3.03	9.63
52	2,600	3.08	3.08	9.77
53	2,650	3.11	3.11	9.86
54	2,700	3.15	3.15	9.97
55	2,750	3.12	3.12	9.88
56	2,800	3.21	3.21	10.13
57	2,850	3.24	3.24	10.21
58	2,900	3.26	3.26	10.26
59	2,950	3.3	3.30	10.37
60	3,000	3.38	3.38	10.58
61	3,050	3.4	3.40	10.63
62	3,100	3.44	3.44	10.73
63	3,150	3.47	3.47	10.81
64	3,160	3.48	3.48	10.83
65	3,170	3.49	3.49	10.86
66	3,180	3.51	3.51	10.91
67	3,190	3.52	3.52	10.93
68	3,200	3.52	3.52	10.93
69	3,210	3.53	3.53	10.96
70	3,220	3.53	3.53	10.96
71	3,230	3.54	3.54	10.98

72	3,240	3.55	3.55	11.00
73	3,250	3.56	3.56	11.03
74	3,300	3.57	3.57	11.05
75	3,350	3.61	3.61	11.15
76	3,400	3.62	3.62	11.17
77	3,450	3.65	3.65	11.25
78	3,500	3.66	3.66	11.27
79	3,550	3.68	3.68	11.32
80	3,600	3.72	3.72	11.41
81	3,650	3.74	3.74	11.46
82	3,700	3.76	3.76	11.50
83	3,750	3.79	3.79	11.57
84	3,800	3.81	3.81	11.62
85	3,850	3.82	3.82	11.64
86	3,900	3.85	3.85	11.71
87	3,950	3.88	3.88	11.78
88	4,000	3.88	3.88	11.78
89	4,100	3.92	3.92	11.87
90	4,200	3.96	3.96	11.95
91	4,300	3.98	3.98	12.00
92	4,400	4.03	4.03	12.11
93	4,500	4.05	4.05	12.15
94	4,600	4.1	4.10	12.26
95	4,700	4.13	4.13	12.32
96	4,800	4.14	4.14	12.34
97	4,900	4.19	4.19	12.44
98	5,000	4.22	4.22	12.51
99	5,500	4.33	4.33	12.73
100	6,000	4.44	4.44	12.95
101	6,500	4.52	4.52	13.10
102	7,000	4.59	4.59	13.24
103	7,500	4.66	4.66	13.37
104	8,000	4.68	4.68	13.40
105	8,500	4.73	4.73	13.50
106	9,000	4.78	4.78	13.59
107	9,500	4.82	4.82	13.66
108	10,000	4.84	4.84	13.70
109	11,000	4.88	4.88	13.77
110	12,000	4.91	4.91	13.82
111	13,000	4.92	4.92	13.84

112	14,000	4.95	4.95	13.89
113	15,000	4.98	4.98	13.94
114	16,000	4.99	4.99	13.96
115	17,000	5	5.00	13.98
116	18,000	5.01	5.01	14.00
117	19,000	5.01	5.01	14.00
118	20,000	5.01	5.01	14.00
119	25,000	5.01	5.01	14.00
120	30,000	5.02	5.02	14.01
121	35,000	4.98	4.98	13.94
122	36,000	4.95	4.95	13.89
123	37,000	4.94	4.94	13.87
124	38,000	4.92	4.92	13.84
125	39,000	4.91	4.91	13.82
126	40,000	4.88	4.88	13.77
127	45,000	4.7	4.70	13.44
128	50,000	4.47	4.47	13.01
129	55,000	4.21	4.21	12.49
130	60,000	3.92	3.92	11.87
131	70,000	3.47	3.47	10.81
132	80,000	3.03	3.03	9.63
133	90,000	2.71	2.71	8.66
134	1,00,000	2.47	2.47	7.85
135	1,50,000	1.68	1.68	4.51
136	2,00,000	1.26	1.26	2.01
137	2,50,000	0.98	0.98	-0.18
138	3,00,000	0.8	0.80	-1.94

Frequency Response:



Result and Discussion:

The frequency response of the first-order active inverting high-pass filter was obtained experimentally. At low frequencies, the gain is strongly attenuated, and it increases gradually as the frequency increases. The experimental cutoff frequency is approximately **3243.9 Hz**, where the gain is about **-3 dB**. At high frequencies, the gain approaches approximately **0 dB**, indicating that the signal is passed with nearly constant gain.

An **unusual behavior** is observed at the lower end of the frequency range, where the gain remains nearly constant around **-38 dB** before the expected rise begins. In addition, the measured roll-off rate is approximately **+26.6 dB/decade**, which is higher than the theoretical **+20 dB/decade** for a first-order high-pass filter. This deviation may be due to the limited low-frequency measurement range, component tolerances, measurement uncertainty, and practical limitations of the op-amp.

Overall, the graph exhibits the expected **high-pass filter characteristic**, with strong attenuation at low frequencies and nearly constant gain at high frequencies.

Experiment 6: Second Order Active High Pass Filter (HPF)

Aim : Design a Non Inverting Second-Order Active High-Pass Filter and study its frequency response.

Given Parameters:

Gain $A_f = 7$ and Cut off Frequency (f_c) = 3.2 kHz

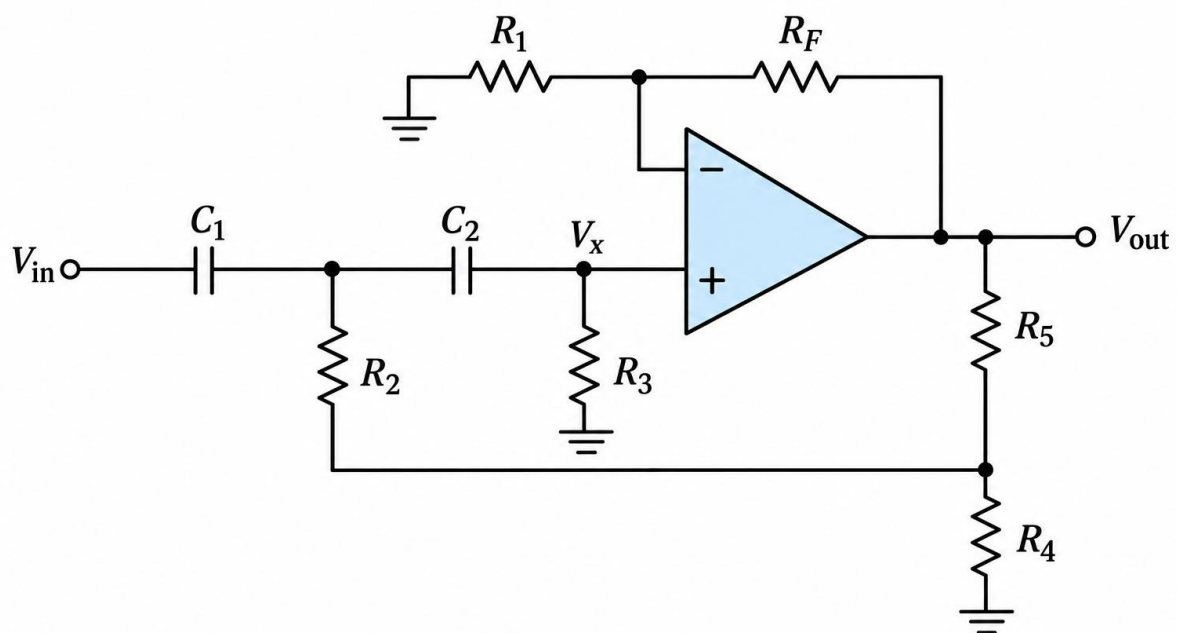
Apparatus Required:

IC 741, Dual track DC power supply ($\pm 12V$ or $\pm 15V$), Function Generator (Sine wave, 10 Hz - 100 kHz), Digital Storage Oscilloscope (DSO) / CRO with probes, Breadboard, Multimeter and single-strand connecting wires

#####Resistors: $R = 2.26\text{ k}\Omega$, $R_f = (1.47\text{ k}\Omega + 9.77\text{ k}\Omega \approx 11.24\text{ k}\Omega)$

#####Capacitor: $C = 0.022\text{ }\mu\text{F}$ or 22 nF

Circuit Diagram:



2nd Order Non-Inverting High-Pass Filter

Theory:

A second-order active High-Pass Filter (HPF) allows frequencies above the cutoff frequency f_c to pass while attenuating frequencies below f_c . The fundamental concepts of a second-order filter, including natural frequency ω_n , quality factor Q , damping ratio ζ , Butterworth response, passband gain, and Sallen-Key operation, are the same as discussed in the Second-Order Active Low-Pass Filter section. The main difference is in the frequency-dependent numerator of the transfer function.

$$H(s) = \frac{Ks^2}{s^2 + \left(\frac{\omega_n}{Q}\right)s + \omega_n^2}$$

Equal-Component Sallen-Key High-Pass Filter

$$R_1 = R_2 = R, \quad C_1 = C_2 = C$$

The cutoff frequency is therefore:

$$f_c = \frac{1}{2\pi RC}$$

$$Q = \frac{1}{3 - K}$$

The behavior of the HPF is opposite to that of the LPF:

- ❖ At very low frequency, the output is strongly attenuated.
- ❖ As frequency approaches f_c , the output magnitude increases.
- ❖ Above f_c , the signal enters the passband.
- ❖ At sufficiently high frequency, the gain approaches the passband gain K .

For a second-order HPF, the asymptotic slope below the cutoff frequency is:

$$+40\text{dB/decade} \quad \text{or} \quad +12\text{dB/octave}$$

High-Gain Modification ($K > 3$)

$$Q = \frac{1}{3 - K}$$

The feedback attenuation factor is:

$$x = \frac{R_4}{R_4 + R_5}$$

and the modified HPF transfer function becomes:

$$H(s) = \frac{K(sRC)^2}{(sRC)^2 + (3 - xK)sRC + 1}$$

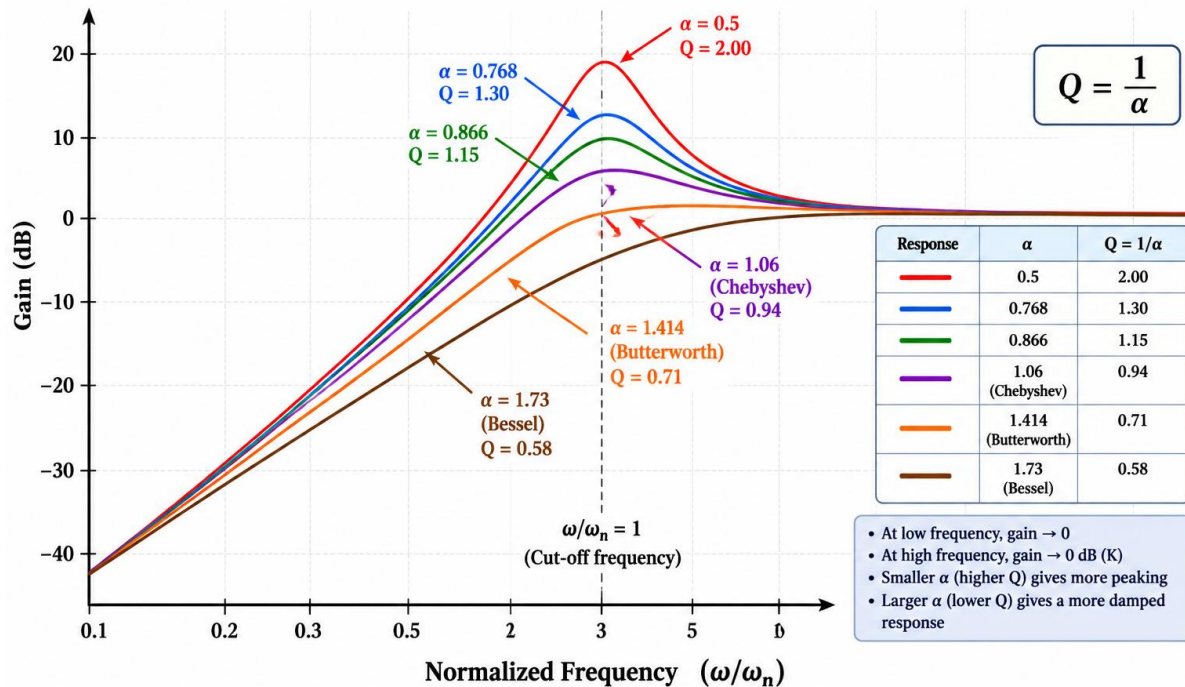
Therefore,

$$Q = \frac{1}{3 - xK}$$

For a Butterworth response:

$$xK = 3 - \sqrt{2} \approx 1.586$$

Second-Order High-Pass Filter Response for Different Damping Coefficient (α) (Quality Factor $Q = 1/\alpha$)



Observations:

#####Resistors: $R = 2.26 \text{ k}\Omega$, $R_f = (1.47 \text{ k}\Omega + 9.77 \text{ k}\Omega \approx 11.24 \text{ k}\Omega)$, Capacitor: $C = 22 \text{ nF}$

$$f_c = \frac{1}{2\pi(2.26 \times 10^3)(22 \times 10^{-9})} \approx 3.20 \text{ kHz}, \quad A_f = -\frac{11.24}{2.26} \approx -4.97$$

$$G_{dB} = 20 \log_{10}(4.97) \approx 13.93 \text{ dB}$$

Experiment 7: Non-Inverting Active Band Pass Filter –

Aim : Design a Non-Inverting Active Band-Pass Filter (BPF) and study its frequency response.

Given Parameters:

Gain $A_f = 7$ and Cut off Frequencies (f_c) = 1 kHz to 4 kHz

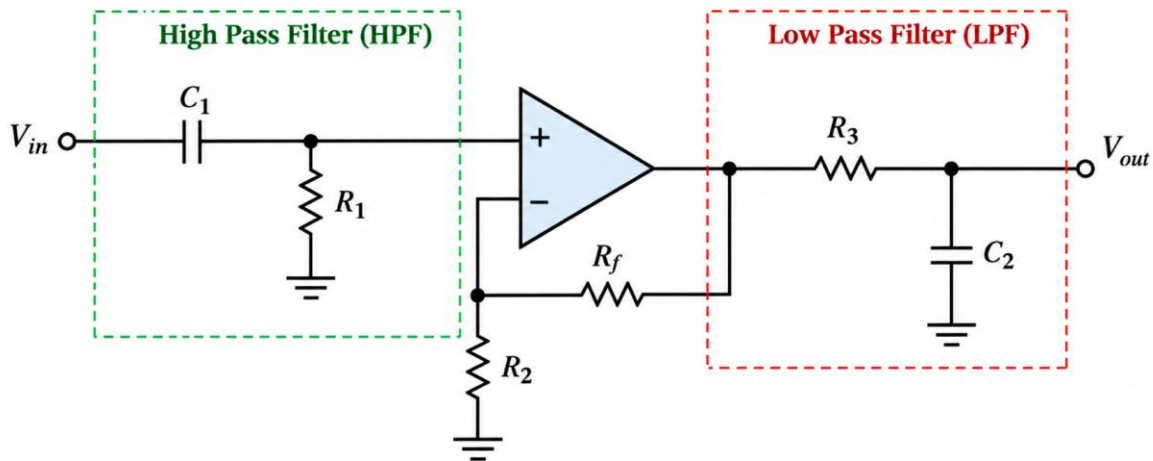
Apparatus Required:

IC 741, Dual track DC power supply ($\pm 12V$ or $\pm 15V$), Function Generator (Sine wave, 10 Hz - 100 kHz), Digital Storage Oscilloscope (DSO) / CRO with probes, Breadboard, Multimeter and single-strand connecting wires

Resistors: $R_1 = 1.47 \text{ k}\Omega$, $R_2 = 3.84 \text{ k}\Omega$, $R_3 = 977 \text{ }\Omega$, $R_f = (384 \text{ }\Omega + 5.53 \text{ k}\Omega)$

Capacitors: $C_1 = 100.8 \text{ nF}$, $C_2 = 10.65 \text{ nF}$

Circuit Diagram:



Theory:

An active band-pass filter allows a **specific range of frequencies** to pass while attenuating frequencies below and above this range. In this circuit, the op-amp is connected in the **non-inverting configuration**, so the output is in phase with the input signal within the passband. The circuit combines **high-pass and low-pass sections**. The $R_1 C_1$ section determines the **lower**

cutoff frequency (f_L), while the R_2C_2 section determines the **upper cutoff frequency (f_H)**. Frequencies between these two cutoff frequencies are passed and amplified by the op-amp.

The resistors R_3 and R_f form the negative-feedback network and determine the passband gain. The use of an op-amp provides voltage gain and buffering, reducing loading effects between the filter and the next stage.

For the non-inverting amplifier, the passband gain is-

$$A_f = 1 + \frac{R_f}{R_3}$$

$$H(s) = \frac{K \left(\frac{\omega_n}{Q} \right) s}{s^2 + \left(\frac{\omega_n}{Q} \right) s + \omega_n^2}$$

The lower and upper cutoff frequencies are approximately determined by the respective RC sections:

$$f_L = \frac{1}{2\pi R_1 C_1}, \quad f_H = \frac{1}{2\pi R_2 C_2}$$

The band-pass filter has:

$$f_L < f_0 < f_H$$

where:

- f_L = lower cutoff frequency
- f_H = upper cutoff frequency
- f_0 = center frequency

The bandwidth is:

$$BW = f_H - f_L$$

The quality factor is:

$$Q = \frac{f_0}{BW}, \quad Q = \frac{f_0}{f_H - f_L}$$

For a second-order band-pass filter, the magnitude response has approximately:

+20 dB/decade slope below the lower cutoff frequency, Maximum response near f_0 and **-20 dB/decade** slope above the upper cutoff frequency.

Gain in dB is given by:

$$A_v(dB) = 20 \log_{10} \left| \frac{V_{out}}{V_{in}} \right|$$

Observations:

Resistors: $R_1 = 1.47 \text{ k}\Omega$, $R_2 = 3.84 \text{ k}\Omega$, $R_3 = 977 \text{ }\Omega$, $R_f = (384 \text{ }\Omega + 5.53 \text{ k}\Omega)$

Capacitors: $C_1 = 100.8 \text{ nF}$, $C_2 = 10.65 \text{ nF}$

$$A_f = 1 + \frac{5.914 \times 10^3}{977} \approx 7.05, \quad G_{dB} = 20 \log_{10}(7.05) \approx \mathbf{16.96 \text{ dB}}$$

The lower cutoff is determined by R_1 and C_1 :

$$f_L = \frac{1}{2\pi(1.47 \times 10^3)(100.8 \times 10^{-9})} \approx \mathbf{1.074 \text{ kHz}}$$

The upper cutoff is determined by R_2 and C_2 :

$$f_H = \frac{1}{2\pi(3.84 \times 10^3)(10.65 \times 10^{-9})} \approx \mathbf{3.895 \text{ kHz}}$$

For a band-pass filter, the centre frequency is approximately the geometric mean of the two cutoff frequencies:

$$f_c = \sqrt{f_L f_H}$$

$$f_c = \sqrt{(1074)(3895)} \approx \mathbf{2.046 \text{ kHz}}$$

Input voltage : 1 Volt

Sr. No.	Frequency (Hz)	Output Voltage (V)	Gain (V_o/V_{in})	Gain(dB)= $20\log_{10}(V_o/V_{in})$
1	50	0.10	0.10	-20.00
2	100	0.78	0.78	-2.16
3	150	1.14	1.14	1.14
4	200	1.49	1.49	3.46
5	250	1.84	1.84	5.30
6	300	2.16	2.16	6.69
7	350	2.47	2.47	7.85
8	400	2.75	2.75	8.79
9	450	3.00	3.00	9.54

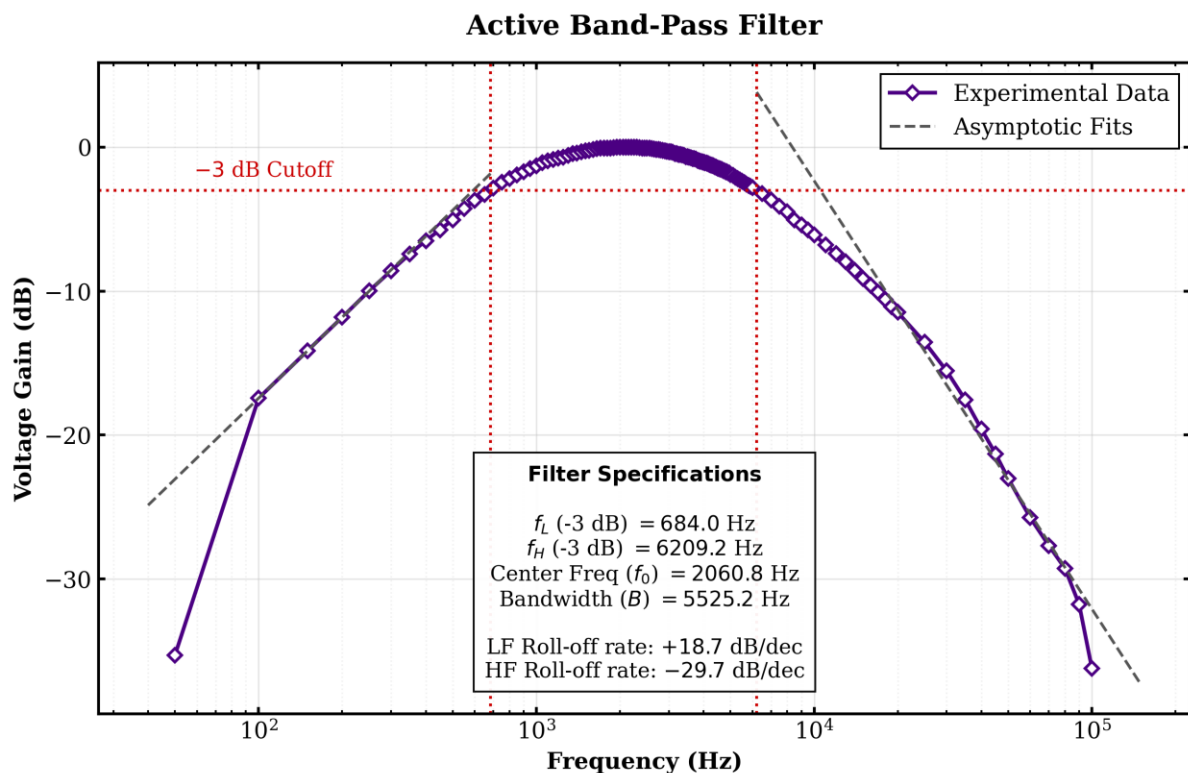
10	500	3.25	3.25	10.24
11	550	3.55	3.55	11.00
12	600	3.79	3.79	11.57
13	650	3.97	3.97	11.98
14	700	4.18	4.18	12.42
15	750	4.38	4.38	12.83
16	800	4.52	4.52	13.10
17	850	4.65	4.65	13.35
18	900	4.77	4.77	13.57
19	950	4.89	4.89	13.79
20	1,000	5.00	5.00	13.98
21	1,050	5.12	5.12	14.19
22	1,100	5.20	5.20	14.32
23	1,150	5.26	5.26	14.42
24	1,200	5.31	5.31	14.50
25	1,250	5.36	5.36	14.58
26	1,300	5.44	5.44	14.71
27	1,350	5.51	5.51	14.82
28	1,400	5.53	5.53	14.85
29	1,450	5.59	5.59	14.95
30	1,500	5.63	5.63	15.01
31	1,550	5.66	5.66	15.06
32	1,600	5.69	5.69	15.10
33	1,650	5.73	5.73	15.16
34	1,700	5.75	5.75	15.19
35	1,750	5.75	5.75	15.19
36	1,800	5.77	5.77	15.22
37	1,850	5.77	5.77	15.22
38	1,900	5.79	5.79	15.25
39	1,950	5.80	5.80	15.27
40	2,000	5.80	5.80	15.27
41	2,050	5.81	5.81	15.28
42	2,100	5.81	5.81	15.28
43	2,150	5.81	5.81	15.28
44	2,200	5.80	5.80	15.27
45	2,250	5.80	5.80	15.27
46	2,300	5.80	5.80	15.27
47	2,350	5.79	5.79	15.25
48	2,400	5.77	5.77	15.22
49	2,450	5.76	5.76	15.21

50	2,500	5.76	5.76	15.21
51	2,550	5.75	5.75	15.19
52	2,600	5.74	5.74	15.18
53	2,650	5.71	5.71	15.13
54	2,700	5.70	5.70	15.12
55	2,750	5.69	5.69	15.10
56	2,800	5.67	5.67	15.07
57	2,850	5.65	5.65	15.04
58	2,900	5.63	5.63	15.01
59	2,950	5.62	5.62	14.99
60	3,000	5.60	5.60	14.96
61	3,050	5.59	5.59	14.95
62	3,100	5.58	5.58	14.93
63	3,150	5.52	5.52	14.84
64	3,200	5.51	5.51	14.82
65	3,250	5.48	5.48	14.78
66	3,300	5.48	5.48	14.78
67	3,350	5.46	5.46	14.74
68	3,400	5.43	5.43	14.70
69	3,450	5.38	5.38	14.62
70	3,500	5.37	5.37	14.60
71	3,550	5.36	5.36	14.58
72	3,600	5.33	5.33	14.53
73	3,650	5.30	5.30	14.49
74	3,700	5.29	5.29	14.47
75	3,750	5.27	5.27	14.44
76	3,800	5.23	5.23	14.37
77	3,850	5.20	5.20	14.32
78	3,900	5.19	5.19	14.30
79	3,950	5.17	5.17	14.27
80	4,000	5.14	5.14	14.22
81	4,050	5.12	5.12	14.19
82	4,100	5.11	5.11	14.17
83	4,150	5.09	5.09	14.13
84	4,200	5.05	5.05	14.07
85	4,250	5.04	5.04	14.05
86	4,300	5.02	5.02	14.01
87	4,350	5.00	5.00	13.98
88	4,400	4.97	4.97	13.93
89	4,450	4.95	4.95	13.89

90	4,500	4.91	4.91	13.82
91	4,550	4.89	4.89	13.79
92	4,600	4.88	4.88	13.77
93	4,650	4.85	4.85	13.71
94	4,700	4.82	4.82	13.66
95	4,750	4.80	4.80	13.62
96	4,800	4.76	4.76	13.55
97	4850	4.76	4.76	13.55
98	4900	4.73	4.73	13.50
99	4950	4.71	4.71	13.46
100	5000	4.67	4.67	13.39
101	5100	4.64	4.64	13.33
102	5200	4.57	4.57	13.20
103	5300	4.52	4.52	13.10
104	5400	4.47	4.47	13.01
105	5500	4.42	4.42	12.91
106	5600	4.37	4.37	12.81
107	5700	4.33	4.33	12.73
108	5800	4.27	4.27	12.61
109	5900	4.24	4.24	12.55
110	6000	4.20	4.20	12.46
111	6500	4.00	4.00	12.04
112	7000	3.81	3.81	11.62
113	7500	3.62	3.62	11.17
114	8000	3.46	3.46	10.78
115	8500	3.25	3.25	10.24
116	9,000	3.12	3.12	9.88
117	9,500	3.00	3.00	9.54
118	10,000	2.88	2.88	9.19
119	11,000	2.66	2.66	8.50
120	12,000	2.48	2.48	7.89
121	13,000	2.32	2.32	7.31
122	14,000	2.17	2.17	6.73
123	15,000	2.03	2.03	6.15
124	16,000	1.92	1.92	5.67
125	17,000	1.82	1.82	5.20
126	18,000	1.72	1.72	4.71
127	19,000	1.62	1.62	4.19
128	20,000	1.55	1.55	3.81
129	25,000	1.22	1.22	1.73

130	30,000	0.97	0.97	-0.26
131	35,000	0.77	0.77	-2.27
132	40,000	0.61	0.61	-4.29
133	45,000	0.50	0.50	-6.02
134	50,000	0.41	0.41	-7.74
135	60,000	0.30	0.30	-10.46
136	70,000	0.24	0.24	-12.40
137	80,000	0.20	0.20	-13.98
138	90,000	0.15	0.15	-16.48
139	1,00,000	0.09	0.09	-20.92

Frequency Response:



Result and Discussion:

The frequency response of the active band-pass filter was obtained experimentally. The filter passes frequencies around its center frequency while attenuating frequencies below and above the passband. The experimental lower and upper cutoff frequencies were found to be **684.0 Hz** and **6209.2 Hz**, respectively, with a center frequency of **2060.8 Hz**. Although the experimental center frequency is close to the specified value of **2 kHz**, the measured cutoff frequencies show considerable deviation from the required **1-4 kHz** range. Consequently, the experimental

bandwidth of **5525.2 Hz** is higher than the specified bandwidth of **3 kHz**. The deviation may be due to component tolerances, loading effects between stages, op-amp limitations. Further, this filter can be designed to achieve a bandwidth of **3 kHz** using two **LM741 ICs**, one for the LPF and the other for the HPF. In this experiment, the measured roll-off rates were approximately **+18.7 dB/decade** at low frequencies and **-29.7 dB/decade** at high frequencies.

Experiment 8: Active Band Reject Filter -

Aim : Design a wide band inverting Active Band-reject Filter (BRF) and study its frequency response.

Given Parameters:

Gain $A_f = 5$ and Cut off Frequencies (f_c) = 1 kHz to 4 kHz

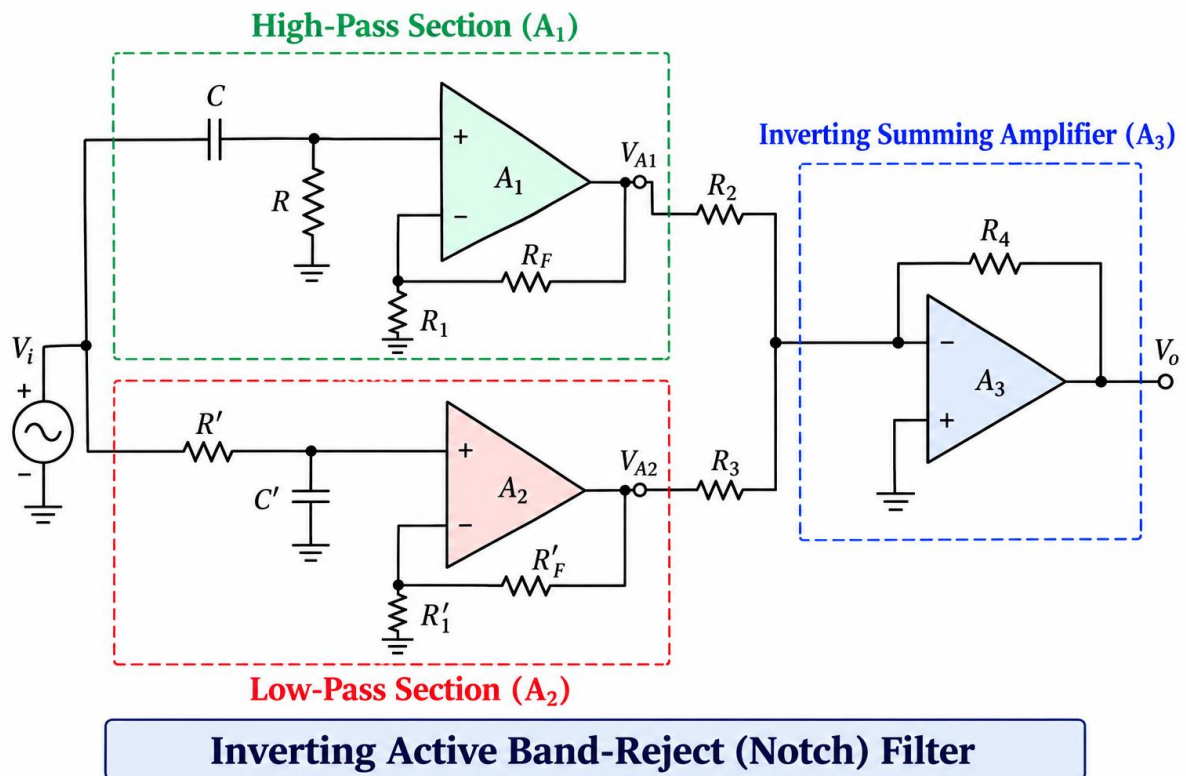
Apparatus Required:

IC 741, Dual track DC power supply ($\pm 12V$ or $\pm 15V$), Function Generator (Sine wave, 10 Hz - 100 kHz), Digital Storage Oscilloscope (DSO) / CRO with probes, Breadboard, Multimeter and single-strand connecting wires

****Resistors:** $R_1 = 1.47 \text{ k}\Omega$, $R_2 = 3.84 \text{ k}\Omega$, $R_3 = 977 \text{ }\Omega$, $R_f = (384 \text{ }\Omega + 5.53 \text{ k}\Omega)$

****Capacitors:** $C_1 = 100.8 \text{ nF}$, $C_2 = 10.65 \text{ nF}$

Circuit Diagram:



Theory:

A second-order active band-reject filter (BRF), also called a band-stop filter, rejects a particular range of frequencies while allowing frequencies below and above that range to pass.

It is therefore the opposite of a band-pass filter in terms of frequency selection.

The filter has two cutoff frequencies:

- Lower cutoff frequency, f_L
- Upper cutoff frequency, f_H

The frequencies between f_L and f_H are attenuated, while frequencies outside this range are passed.

High-pass section -A₁

The input signal is applied to the **high-pass section** containing the capacitor and resistor.

- ❖ At **low frequencies**, the capacitor opposes the signal, so very little signal reaches A₁.
- ❖ At **high frequencies**, the capacitor allows the signal to pass easily, so A₁ produces a strong output. Therefore, A₁ mainly provides the **high-frequency components**.

Low-pass section- A₂

The same input signal is applied to the **low-pass section** containing the resistor and capacitor.

- ❖ At **low frequencies**, the signal passes through this section and A₂ produces a strong output.
- ❖ At **high frequencies**, the capacitor diverts the high-frequency signal toward ground, so A₂ output becomes small. Therefore, A₂ mainly provides the **low-frequency components**.

Combining the two sections- A₃

The outputs of A₁ and A₂ are fed to the **inverting summing amplifier A₃** through R₂ and R₃. A₃ combines the signals from both paths and produces the final output.

The standard second-order band-reject transfer function is:

$$H(s) = K \frac{s^2 + \omega_n^2}{s^2 + \left(\frac{\omega_n}{Q}\right)s + \omega_n^2}$$

At the rejection frequency,

$$K_1 = 1 + \frac{R_F}{R_1}$$

$$K_2 = 1 + \frac{R_F'}{R_1'}$$

The outputs of the HP and LP sections are combined at **A₃**.

$$V_o = -R_4 \left(\frac{V_{A1}}{R_2} + \frac{V_{A2}}{R_3} \right)$$

The frequency response consists of three main regions:

Below f_L : Low-frequency signals are passed with approximately the passband gain.

Between f_L and f_H : The selected frequency band is attenuated. The maximum attenuation occurs near f_0 .

Above f_H : High-frequency signals are again passed with approximately the passband gain.

Therefore, the frequency response has a **notch-shaped characteristic**.

Notch Frequency

At the center frequency $f = f_0$ the filter provides maximum rejection.

Ideally, for the standard second-order band-reject transfer function:

$$H(j\omega_0) = 0$$

This means that the output ideally becomes zero at the notch frequency. In a practical circuit, complete cancellation may not occur because of resistor and capacitor tolerances, op-amp limitations, and component mismatch.

Gain in dB is given by,

$$A_v(dB) = 20 \log_{10} \left| \frac{V_{out}}{V_{in}} \right|$$

Observations:

#####Resistors: $R_1 = 1.47 \text{ k}\Omega$, $R_2 = 3.84 \text{ k}\Omega$, $R_3 = 977 \text{ }\Omega$, $R_f = (384 \text{ }\Omega + 5.53 \text{ k}\Omega)$

Capacitors: $C_1 = 100.8 \text{ nF}$, $C_2 = 10.65 \text{ nF}$

$$A_f = 1 + \frac{5.914 \times 10^3}{977} \approx 7.05, \quad G_{dB} = 20 \log_{10}(7.05) \approx 16.96 \text{ dB}$$

The lower cutoff is determined by R_1 and C_1 :

$$f_L = \frac{1}{2\pi(1.47 \times 10^3)(100.8 \times 10^{-9})} \approx 1.074 \text{ kHz}$$

The upper cutoff is determined by R_2 and C_2 :

$$f_H = \frac{1}{2\pi(3.84 \times 10^3)(10.65 \times 10^{-9})} \approx 3.895 \text{ kHz}$$

For a band-pass filter, the centre frequency is approximately the geometric mean of the two cutoff frequencies:

$$f_c = \sqrt{f_L f_H}$$

$$f_c = \sqrt{(1074)(3895)} \approx 2.046 \text{ kHz}$$

